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SECTION 562-2
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MILITARY STANDARD

DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS

SECTION 562-2

CONTROL SURFACE STRUCTURE DESIGN

AMSC N/A FSC 1990

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FOREWORD

1. This military standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use within the distribution limitations noted on the cover page.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. The purpose of this standard is to set forth design methods for control surface structures, such as rudders and diving planes, for naval surface ships and submarines.

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1. GENERAL AND SCOPE

1.1 General. The procedures established by MIL-STD-2189(SH) are applicable. This section and the basic standard are to be considered as an integral single document.

1.2 Scope. This standard is issued to provide a design procedure for control surface structure design. The procedure is applicable to surface ships and submarines, and applies to rudders, fin stabilizers, stern diving planes, fairwater planes, and bow planes. The structure considered consists of the control surface, its stock, web spacing, web and plating thickness, hub, gudgeons, pintles, keyways, and bearing configuration. The shock design of control surfaces is not considered herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents. This paragraph is not applicable to this standard.

2.2 Non-Government publication. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the Department of Defense Index of Specifications and Standards (DODISS) are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN BUREAU OF SHIPPING (ABS)

Rules for Building and Classing Steel Vessels (annual)

(Application for copies should be addressed to the American Bureau of Shipping, 45 Eisenhower Drive, P.O. Box 910, Paramus, NJ 07653-0910.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions. Unless otherwise defined herein, the terms used have their usual dictionary meaning appropriate to the context.

3.1.1 Diving Planes. The diving planes are the control surfaces used for controlling the vertical motion of a submarine. Diving planes include stern diving planes, fairwater planes, and bow planes.

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3.1.2 Fairwater planes and bow planes. Fairwater planes and bow planes are the control surfaces built into the submarine fairwater (sail) or near the bow, used for controlling the vertical motion of a submarine. Fairwater planes and bow planes have a configuration similar to that of spade rudders (see figure 1).

3.1.3 Fin stabilizers. Fin stabilizers are the control surfaces used for reducing surface ship roll. Fin stabilizers are similar in configuration to spade rudders (see figure 1).

3.1.4 Gudgeon. The gudgeon is the socket on a horn rudder or balanced rudder into which the pintle fits. It's normally part of the rudder (see figures 2 and 3); similarly for stern diving planes.

3.1.5 Hub. The hub is the casting at the root of a control surface which mates with the stock (see figure 1).

3.1.6 Keyway. The keyway is the matching slot in the hub and stock of a spade or horn rudder for receipt of a torque key. Torque keys constrain the rudder and stock to rotate together; similarly for diving planes and fin stabilizers.

3.1.7 Pintle. The pintle is the pin at the bottom of a balanced rudder which mates with the gudgeon (see figure 3).

3.1.8 Rudder. The rudder is the control surface used for control of the horizontal motion of a surface ship or submarine. Three types of rudders are used in Naval ship design practice: Spade, horn, and balanced (see figures 1, 2, and 3).

3.1.9 Stern diving planes. The stern diving planes are the control surfaces located near the stern of a submarine for influencing the vertical motion, primarily by controlling the pitch angle. Stern diving planes have a configuration similar to that of balanced rudders except that they are usually located aft of fixed horizontal stabilizers.

3.1.10 Stock. The stock is the rotating element that articulates the control surface. The control surface is locked to the stock by a system of keys. The stock is rotated by the steering gear machinery located in the ship. The stock passes through the hull of the ship by way of a system of glands and seals (see figures 1, 2, and 3); similarly for other control surfaces.

3.2 List of symbols and abbreviations.

a = Length of panel

a = Load at tip of span, or at aft end of chord

a = One-tenth mean radius of gudgeon = $(R_1 + R_2)/20$

A = Projected area of control surface

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A = Designation of point on stock centerline at center of upper bearing; also the x-coordinate of this point

A_b = Area of horn-type control surface between pintle and root

A_c = Area of horn-type control surface between pintle and tip

A_h = Area in horizontal section enclosed by horn

b = Breadth of panel

b = Load at root of span or at forward end of chord

B = Span

B = Designation of point on stock centerline at center of lower bearing or neck bearing; also the x-coordinate of this point

c = Auxiliary variable for use in calculating hub casting wall thickness; $c = s + d/2$

c = Mean chord length

C = Constant depending on material (see 5.6.1.1)

℄ = Centerline

CP = Center of pressure

d = Diameter of annular groove in stock

d = Stock diameter

d = Minimum stock diameter

d_1 = The first of two tentatively determined stock diameters

d_2 = The second of two tentatively determined stock diameters

d_i = Inner diameter of stock

d_o = Outer diameter of stock

E = Modulus of elasticity

F = Rudder force, proportional to area of load trapezoid

F = Tension force on stock

G = Gudgeon strength factor (see figure 17)

h_a = Distance from center of neck bearing to root of control surface (see figure 14)

h_d = Distance from center of pintle bearing of balanced control surface to tip (see figure 14)

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h_n = Distance from center of lower bearing to root of control surface in spade or balanced configuration, or distance from center of neck bearing to tangent line to hull where horn joins hull in horn configuration (see figures 10, 12, and 14)

h_r = Length of span of balanced control surface (see figure 14)

H = Depth of keyway

H = Maximum total head of seawater

HS = Higher strength (said of steel)

HY = Higher yield strength (said of steel)

I = Second moment of area of section of stock about a diameter (commonly called moment of inertia of section)

I_a = Mean moment of inertia of stock

I_b = Mean moment of inertia of control surface inboard from pintle bearing

I_d = Mean moment of inertia of sole piece (also called shoe)

I_h = Mean moment of inertia of horn

I_H = Moment of inertia of hub and associated stock to be used in strength calculations

I_{H-A} = Actual moment of inertia of hub and associated stock about centroidal axis of control surface section

J_h = Mean polar moment of inertia of horn

k = Horn spring constant, force per unit length

k_d = Shoe (also called sole piece) spring constant, force per unit length

K = Constant depending on b/a (see 5.6.1.1)

K = Stress factor (dimensionless) depending on stock penetration (see figure 16)

K_I = Stress concentration factor at fillet (see figure 19)

kip = 1000 pounds

L = Length of gudgeon

L = Length of cord or span

L_a = Length of stock between upper and lower bearing centers (see figures 10, 12, and 14)

L_b = Distance from lower bearing center to tip of control surface or to pintle bearing center (see figures 10, 12, and 14)

L_c = Distance from center of pintle bearing for horn control surface to tip edge (see figure 12)

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L_d = Distance from centerline of pintle to assumed point of fixity of sole piece (also called shoe) to hull or other structure (see figure 14)

L_F = Distance from stock centerline to line of action of resultant of forces acting on control surface forward of stock centerline (see figure 11)

L_h = Distance along stock centerline from center of pintle bearing to root of control surface (see figures 12 and 13)

L_R = Distance from stock centerline to line of action of resultant of forces acting on control surface aft of stock centerline (see figure 11)

L_r = Root chord length (see figure 10)

L_t = Tip chord length (see figure 10)

m = Distance from neutral axis of horn to center of pintle (see figure 12)

M = Bending moment in stock

M_F = Maximum bending moment acting at stock centerline due to forces on end of control surface forward of stock centerline (see figure 14)

M_h = Bending moment at neck bearing or lower bearing (see figures 10, 12, and 14)

M_p = Bending moment at pintle bearing (see figure 12)

M_R = Maximum bending moment acting at stock centerline due to forces on end of control surface aft of stock centerline (see figure 11)

M_t = Torque in stock

M_x = Bending moment at point X, a distance x from point of origin

n = Factor of safety on stock stresses

N = Resultant force acting at CP (see figure 10)

N_1 = Total force on control surface for horn and balanced rudders

N_F = Resultant of chordwise forces acting on control surface forward of stock centerline (see figure 11)

N_R = Resultant of chordwise forces acting on control surface aft of stock centerline (see figure 11)

OS = Ordinary strength (said of steel)

P = Ratio of inner diameter of stock to outer diameter of stock

P_c = Pintle bearing reaction, especially that for balanced control surface (see figure 14)

P_F = N_F

P_h = Lower bearing reaction (see figure 10)

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P_n = Penetration of stock in hub, in stock diameters

P_p = Pintle bearing reaction, especially that for horn-type control surface (see figure 12)

P_R = N_R

P_u = Upper bearing reaction (see figures 10, 12, and 14)

r = Radius of fillet in stock annular groove

r_1 = Radius of fillet in keyway (see figure 18)

R_1 = Outside radius of gudgeon

R_2 = Inside radius of gudgeon

s = Hub wall thickness

s_o = Stress on rudder stock in bending, tension, or torsion

SCF = Stress concentration factor

t = Thickness of outside control surface plating

t_{AB} = The tangential deviation of a point A from a tangent to the elastic curve (of stock) at B

t_{BA} = The tangential deviation of a point B from a tangent to the elastic curve (of stock) at A

t_{Bx} = The tangential deviation of a point B from a tangent to the elastic curve (of stock) at x

t_{xA} = The tangential deviation of a point at x from a tangent to the elastic curve (of stock) at A

t_{xB} = The tangential deviation of a point at x from a tangent to the elastic curve (of stock) at B

t_1 = Corrosion allowance in thickness of plating

t_2 = Cavitation allowance in thickness of plating

t' = Thickness of plating based on loading

t'_1 = Assumed base thickness of outside plating

T = Total reaction at gudgeon (or shear force at horn)

V_x = Deflection of stock at coordinate x

W = Chordwise loading, force per unit length (see figure 11)

W_b = Uniform load over part of control surface between pintle and root, force per unit length (see figure 12)

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W_c = Uniform load over part of control surface between pintle and tip, force per unit length (see figure 12)

W_t = Hydrodynamic pressure

x = Distance of center of pressure from top (root) of span or from forward end of chord

x = Coordinate for locating points along axis of stock

x_s = Coordinate for locating position of seal along axis of stock

z = Distance of center of pressure from root of control surface (see figure 10)

σ = Reduced yield strength of hub material $=0.6\sigma_y$

σ_G = Maximum stress in gudgeon

σ_y = Yield strength of stock material

$\Sigma\frac{s}{t}$ = Circumferential length of control surface horn in horizontal section, divided by its mean wall thickness (dimensionless ratio)

4. GENERAL REQUIREMENTS

4.1 Design flow diagram. The flow diagram shown on figure 4 indicates interrelationships of the various parts of the design procedure. The arrows indicate the sequence of steps that shall normally be followed in developing a design. With control surfaces of the horn type or balanced type, estimated values of stock and control surface moments of inertia are needed in order to compute shear and bending moments. Therefore, determination of stock size, web spacing, and web and side plating thicknesses shall be considered first. The resulting calculations typically form an iterative loop in the flow diagram until the calculated values become mutually compatible.

4.2 Input information. The information needed for carrying out the design calculations shall be as presented in 4.2.1 through 4.2.6.

4.2.1 Bearings. Bearings shall bear the stock radial and thrust loads and accommodate the stock flexural deflections. Two basic types of bearings may be used: Sleeve and antifriction.

4.2.1.1 Sleeve bearings. Sleeve bearings have low initial cost, are relatively easy to maintain, and in most cases are relatively easy to repair in emergencies.

4.2.1.2 Antifriction bearings. Antifriction bearings are fairly expensive, require some precision machining to install, and require a careful sealing arrangement to prevent seawater contamination; they have the compensatory advantages of permitting significant reduction of steering gear torque and possibly being self-aligning (so that stock deflections and hull movement are less likely to cause binding).

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4.2.1.3 Coefficients of friction. The estimated friction coefficients shall be 0.01 for antifriction and 0.20 for sleeve type bearings.

4.2.2 Arrangement and location. Principal control surface internal arrangements and locations of bearings are determined on the basis of previous experience. Five typical arrangements of rudder stocks and bearings are shown on figures 5 through 9. A tabular listing of bearing particulars for each configuration is presented in table I. The sleeve bearing length (or height) shall be not less than 1.2 times the diameter of the stock. The separation of the upper and lower bearings shall be not less than one-third the distance from the center of the lower bearing to the control surface spanwise center of pressure. This requirement keeps the required stock diameter within reasonable limits.

TABLE I. *Control surface bearing configuration particulars.*

Arrangement type designator	Loading function	Upper bearing		Lower (or neck) bearing	
		Type of bearing	Lubricant	Type of bearing	Lubricant
A (figure 5)	Radial	Sleeve	Grease	Sleeve	Water and grease
B (figure 6)	Radial	Sleeve	Grease	Sleeve	Water
C (figure 7)	Carrier and radial	Antifriction	Grease	Sleeve	Grease
D (figure 8)	Carrier and radial	Antifriction	Grease	Antifriction	Grease
E (figure 9)	Carrier and radial	Antifriction	Grease	Antifriction	Grease

4.2.3 Determination of total forces and moments acting on control surface structure. Hydrodynamic, sea slap, hydrostatic force, and ice load shall be considered in their total combined effect.

4.2.3.1 Hydrodynamic load. The structural design of the diving planes, sternplane stabilizers, rudders, stocks and their associated supports and bearings shall be based on the maximum hydrodynamic loads in ahead operation full design speed.

4.2.3.2 Sea slap. A rectangular shaped spanwise sea slap head distribution may be assumed for ship rudders. The value of the sea slap head shall be not less than 1,000 lb/ft^2.

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4.2.3.3 Hydrostatic force. This pressure is due to the hydrostatic head of water acting on the control surface and shall be calculated from the depth of water at the point on the control surface for which the pressure is calculated. The hydrostatic load does not produce a torque or bending moment on the control surface or stock, but merely induces a pressure on the exterior which must be supported by the plating and the internal structure.

4.2.3.4 Ice load. This load is due to the impact of ice on the control surface. For auxiliary ships, see 5.12.1. For other surface ships, see 5.12.2. For submarines see 5.12.3.

4.2.4 Material yield strength. The minimum yield strength to be used for low carbon alloy steel rudder stocks for auxiliary ships shall be 65,000 lb/in^2, and for combatant ships 100,000 lb/in^2; however, if the rudder root-thickness-to-chord ratio becomes greater than 0.23 (after planform alternatives have been considered) 100,000 lb/in^2 yield point of steel should also be considered for auxiliary ships' stock material. Where rudder stocks are required to have little or no magnetic permeability, nickel aluminum bronze shall be used. The minimum yield strength for nickel aluminum bronze is 35,000 lb/in^2. The minimum yield strength for steel hub castings is 30,000 lb/in^2. Rudder plating shall be made of HY-80 (for combatant ships) or ABS (for auxiliary ships) steel, and internal members shall be made of HS or OS steel.

4.2.5 Factors of safety. For surface ships a factor of safety of not less than 2.0 on yield for spade rudders, nor less than 2.5 on yield for horn and balanced rudders, shall be used. For submarines, a factor of safety of 2.0 on yield for hydrodynamic or ice loads, and a factor of safety of 1.25 on yield for sea slap, shall be used. These factors of safety are to be used in conjunction with the shear stress.

4.2.6 Shear stress limit. The maximum allowable shear stress shall be taken as 60 percent of yield strength.

5. DETAILED REQUIREMENTS

5.1 Determination of shear, bending moment, and deflection diagrams. The given chordwise and spanwise loads shall be combined. These combined loads shall be used for the overall shear and bending moment calculations only.

5.2 Calculation of load distributions over chord and over span. The approach taken is to use the calculated force and the center of hydrodynamic pressure to derive the chordwise and spanwise load distributions under the assumption that they fit a trapezoidal distribution. The technique shall cause the centroid of the load distribution trapezoid to coincide with the spanwise (or chordwise) center of pressure.

The calculation procedure shown below solves equations (1) and (2) for the loads (a and b) at the forward and aft ends of the mean chord and also at the top and bottom (root and tip) of the span:

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$$F = \frac{a + b}{2} L \tag{1}$$

$$x = \frac{L(2a + b)}{3\ (a + b)} \tag{2}$$

where,

a = load at aft end of mean chord or bottom (tip) of span (lb/in)

b = load at forward end of mean chord or top (root) of span (lb/in)

x = distance of center of pressure from forward end of chord or from top (root) of span = abscissa of centroid of load trapezoid (inches)

L = length of chord or span (inches)

F = rudder force, proportional to area of load trapezoid (lb)

Solving equations (1) and (2) gives the spanwise and chordwise load distributions a and b:

$$a = \frac{2F}{L}\left(\frac{3x}{L} - 1\right) \qquad (lb/in)$$

$$b = \frac{2F}{L}\left(2 - \frac{3x}{L}\right) \qquad (lb/in)$$

5.2.1 Spade type control surface, spanwise load. The spanwise loading and the resulting shear and moment diagrams for a spade type control surface are shown on figure 10.

The bearing reaction forces shall be calculated as shown in equations (3) and (4).

$$P_u = \frac{N(z + h_n)}{L_a} \qquad (kip) \tag{3}$$

$$P_h = N + P_u \qquad (kip) \tag{4}$$

The maximum bending moment shall be calculated as shown in equation (5).

$$M_h = N(z + h_n) \tag{5}$$

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In applying equations (3) through (5),

N = Resultant force acting at CP (kip)

P_u = Upper bearing reaction (kip)

P_h = Lower bearing reaction (kip)

CP = Center of hydrodynamic pressure

M_h = Maximum moment at lower bearing (in-kip)

h_n, L_a, L_b, L_r, L_t, and z are lengths shown in inches (see figure 10)

For spade control surfaces partially in the propeller race a resultant CP and N may be determined from the given loads on the portion within the propeller race and that outside the race. Then the same procedure as outlined above can be used.

5.2.2 Spade type control surface, chordwise load. The chordwise loading, shear, and bending moment diagrams are as shown (see figure 11).

The maximum chordwise bending moment shall be calculated as follows:

$$M_R = L_R N_R \quad (6)$$

$$M_F = L_F N_F \quad (7)$$

where,

N_R = Resultant of chordwise forces acting on control surface aft of stock centerline (kip)

N_F = Resultant of chordwise forces acting on control surface forward of stock centerline (kip)

W = Chordwise loading (kip/ft)

P_F = N_F (kip)

P_R = N_R (kip)

M_R = Maximum bending moment acting at stock centerline due to forces on end of control surface aft of stock centerline (in-kip)

M_F = Maximum bending moment acting at stock centerline due to forces on end of control surface forward of stock centerline (in-kip)

L_R = Distance from stock centerline to line of action of resultant of forces acting on control surface aft of stock centerline (inches)

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L_F = Distance from stock centerline to line of action of resultant of forces acting on control surface forward of stock centerline (inches)

5.2.3 Horn type control surfaces, spanwise load. The spanwise load diagram, shear diagram, and bending moment diagram for a horn type control surface are as illustrated (see figure 12). The bearing reaction forces for upper, lower (neck), and pintle bearings shall be calculated as shown in equations (8), (9), and (10), respectively.

$$P_u = \frac{M_h}{L_a} \tag{8}$$

$$P_h = P_u(1 + L_a/L_b) + (W_b/(2\ L_b)\ (L_b - h_n)^2 - M_p/L_b \tag{9}$$

$$P_p = N_1 + P_u - P_h \tag{10}$$

The maximum bending moments are:

$$M_p = \frac{W_c\ L_c^2}{2} \tag{11}$$

$$M_h = \frac{W_b\ (L_b^2 - h_n^2)(\frac{1}{2} + \frac{a}{8}) + W_c L_c^2\ (\frac{1}{2} + \frac{L_b}{L_c} - \frac{a}{4})}{1 + a(1 + \frac{4\ L_a I_b}{3\ L_b I_a})} \tag{12}$$

where,

$$a = \frac{k\ L_b^3}{3EI_b} \tag{13}$$

$$k = \frac{E}{L_h\ (\frac{2.6m^2}{J_h} + \frac{L_h^2}{3I_h})} \qquad \textit{kip/in, the horn spring constant} \tag{14}$$

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W_b = Uniform load over upper part of control surface (kip/in)

W_c = Uniform load over lower part of control surface (kip/in)

I_a = Mean moment of inertia of stock (in^4)

I_b = Mean moment of inertia of control surface inboard from pintle bearing (in^4)

I_h = Mean moment of inertia of horn (in^4)

E = Modulus of elasticity (kip/in^2)

P_u = Upper bearing reaction (kip)

P_h = Lower bearing reaction (kip)

P_p = Pintle bearing reaction (kip)

M_h = Bending moment at neck bearing (in-kip)

M_p = Bending moment at pintle bearing (in-kip)

N_1 = Total force on control surface (kip)

J_h = Mean polar moment of inertia of horn (in^4)

$$= 4\frac{A_h^2}{\Sigma\frac{s}{t}} \quad (in^4)$$

A_h = Area in horizontal section enclosed by the horn (in^2)

$\Sigma\frac{s}{t}$ = Circumferential length of control surface horn in horizontal section, divided by its mean wall thickness, (dimensionless ratio)

h_n, L_a, L_b, L_c, and L_h are lengths as shown on figure 12 (inches)

m = Distance from neutral axis of horn to center of pintle on figure 12 (inches)

For horn control surfaces partially in the propeller race a CP and N may be determined from the given loads on the portion within the propeller race, and that outside the race. Then the same procedure as outlined above can be used.

5.2.4 Control surface horn load. The shear force, bending moment, and torque acting on a horn rudder horn shall be calculated as shown on figure 13.

5.2.5 Horn type control surface, chordwise load. The procedure for a spade type control surface is applicable (see 5.2.2).

5.2.6 Balanced type control surface, spanwise load. The spanwise loading diagram, shear diagram, and bending moment diagram for a balanced type control surface are shown on figure 14.

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5.2.6.1 Bearing reaction. The bearing reaction forces shall be calculated as follows:

$$P_u = \frac{M_h}{L_a} \qquad (kip) \tag{15}$$

$$P_h = P_u \left(1 + \frac{L_a}{L_b}\right) + \frac{N}{L_b}\left(\frac{h_r}{2} + h_d\right) \qquad (kip) \tag{16}$$

$$P_c = N_1 + P_u - P_h \qquad (kip) \tag{17}$$

5.2.6.2 Bending moments. The bending moments shall be calculated as follows:

$$M_h = \frac{W_b \left(L_b^2 - h_n^2\right)\left(\frac{1}{2} + \frac{a}{8}\right)}{1 + a\left(1 + \frac{L_a I_b}{L_b I_a}\right)} \qquad (in\text{-}kip) \tag{18}$$

$$M_F = \frac{W_b \, h_r \, 2}{8} \qquad (in\text{-}kip) \tag{19}$$

5.2.6.3 Alternative neck bearing location. When the neck bearing is located outboard of the control surface root chord and $h_n < 0.1\ h_r$ (see figure 14), equation (18) may be used if the term "$- h_n^2$" in the numerator is replaced by "$+ h_n^2$".

5.2.6.4 Notation summary. In equations (15) through (18),

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$$a = \frac{k_d L_b^3}{3EI_b} \quad OR \quad \frac{L_b^3 I_d}{L_d^3 I_b}$$

$$k_d = \frac{3EI_d}{L_d^3}$$

N_1 = Total force on control surface (kip)

I_a = Mean moment of inertia of stock (in^4)

I_b = Mean moment of inertia of control surface (in^4)

I_d = Mean moment of inertia of sole piece (in^4)

L_d = Distance from centerline of pintle to assumed point of fixity of sole piece to hull or other structure (in)

W_b = Uniformly distributed loading on control surface (kip/in)

M_h = Bending moment at lower bearing (in-kip)

M_F = Maximum bending moment on control surface (in-kip)

P_u = Upper bearing reaction (kip)

P_h = Lower bearing reaction (kip)

P_c = Pintle bearing reaction (kip)

E = Modulus of elasticity (kip/in^2)

h_d, h_r, h_n, L_a, and L_b are lengths as shown on figure 14 (inches).

5.2.7 Balanced type control surface, chordwise loading. Balanced type control surfaces shall be treated the same as for spade type control surfaces (see 5.2.2).

5.3 Shear and bending moment diagrams and bearing reactions. The shear and bending moment diagrams and bearing reactions shall be determined for each combined loading. The following requirements shall apply:

a. The shear curves presented in the foregoing procedures were determined by integrating the spanwise and chordwise load curves over the span and chord respectively. The bending moment curves were determined by integrating the shear curves.

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b. Tiller forces have not been considered in the foregoing procedures. If a tiller force exists, it may be included by determining the shears and bending moments due to the tiller force only, and superimposing them on those outlined herein. Appropriate boundary conditions shall be assumed at the bearings for determining tiller shear forces and bending moments.

c. Mean moments of inertia for horn and balanced control surfaces, horns, and sole pieces may be used in determining bending moments for horn and balanced control surfaces only in normal designs where variations in moments of inertia are not great. In other cases the variation of moments of inertia shall be taken into account.

5.4 Stock deflection analysis. The stock deflection at the seal shall be such that a full load deflection shall not cause excessive leakage through the seal. The moment-area method can be used to find the stock deflection at any desired location. Steps in the procedure are as follows:

a. Tabulate the bending moment M from the upper bearing to the location of the seal. Let x be measured along the stock centerline, with x = A at the upper bearing, x = B at the lower bearing (B > A), and x = x_s at the seal location (see figure 12).

b. Determine the moment of inertia I at each x location (also called an x station) selected in step (a).

c. Calculate M/(EI) at each x station (E is the modulus of elasticity).

For x > B, the deflection is:

$$v_x = t_{xB} + [(x - B)/(B - A)]\, t_{AB}$$

In the foregoing procedure:

t_{xB} = the tangential deviation of a point x from a tangent to the elastic curve at B.

t_{xB} = (the definite integral proportional to the area under the M/(EI) curve from x to B) times (the distance from x to the centroid of that area).

Other deviations are calculated similarly, that is,

t_{AB} = (the definite integral proportional to the area under the M/(EI) curve from A to B) times (the distance from A to the centroid of that area).

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In general, t_{xB} will not be equal to t_{Bx}.

For x < B, the deflection is:

$$v_x = \left(\frac{X}{B - A}\right) t_{BA} - t_{xA}$$

This method is applicable to control surfaces with or without outboard bearings (pintle bearings), and is valid as long as the M/(EI) curve does not cross through zero in the domain A to X_s. If the curve does cross through zero in this domain, the moment area method can still be used, but the foregoing formulas will not be valid and a more detailed moment area calculation will need to be used.

5.5 Determination of stock size. The stock diameter shall be determined by equation (20) or (21). The minimum stock diameter shall be not less than the larger of d_1 or d_2. Stocks may have bores for the purposes of heat treatment and weight reduction.

$$d_1 = \sqrt[3]{\frac{16n}{\sigma_y \pi (1 - p^4)} \left(M + \sqrt{M^2 + M_t^2}\right)} \tag{20}$$

$$d_2 = \sqrt[3]{\frac{16n}{0.6\sigma_y \pi (1 - p^4)} \sqrt{M^2 + M_t^2}} \tag{21}$$

where,

d = Minimum stock diameter, the larger of d_1 or d_2 (inches)

M = Bending moment in stock (in-kip)

M_t = Torque in stock (in-kip)

(For torque: In way of keyways the torque shall be increased as specified in 5.9.3.)

σ_y = Yield strength of stock material (kip/in^2)

n = 2.0 for surface ship spade control surface

2.5 for surface ship horn control surface

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2.0 for submarine control surface (hydrodynamic or ice loads)

1.25 for submarine control surface (sea slap)

P = Ratio of inner diameter of stock to outer diameter of stock (dimensionless)

5.6 Determination of web spacing and web and side plating thickness. Control surfaces shall be constructed with horizontal and vertical webs covered by side plating.

5.6.1 Nonfilled control surfaces. (See 5.6.1.1 through 5.6.1.3 and see 5.6.3.)

5.6.1.1 Ratio of panel breadth to thickness of plating. This ratio shall be based on load as shown in equation (22).

$$\frac{b}{t'} = \frac{C}{\sqrt{K H}} \tag{22}$$

where,

b = Breadth of panel (inches)

a = Length of panel (inches)

The following relationship exists between a and b:

$$b \approx a$$

t′ = Thickness of plating based on loading (inches)

H = Maximum total head of seawater (ft)

= The sum of the hydrostatic head due to hydrodynamic pressure (or ice load) plus the hydrostatic head due to draft of the ship or sea slap, whichever of these two latter addends is greater.

a. *Hydrodynamic head.* The spanwise head distribution may be determined from the given spanwise load distribution. The chordwise distribution of the head may be assumed similar to the given total chordwise distribution at the mean chord.

b. *Ice load.* (See 4.2.3.4.)

c. *Sea slap head.* A rectangular shaped spanwise sea slap head distribution may be assumed for surface ship rudders. The value of the sea slap head shall be not less than 1,000 lb/ft^2 for both surface ships and submarines.

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d. *Hydrostatic head.* The hydrostatic head shall be determined based on the full load draft of the ship.

C = Constant depending on material ($ft^{1/2}$) (see table II).

K = Constant depending on b/a (dimensionless) (see table III).

TABLE II. *Values of "C" for steel types.*

Types of Steel	Value of C ($ft^{1/2}$)
OS	550
HS	620
HY80	750
HY100	810

TABLE III. *Values of "K" as function of b/a.*

b/a	Value of K (dimensionless)
0.5 or less	1.00
0.6	.98
0.7	.94
0.8	.89
0.9	.84
1.0	.78

5.6.1.2 Plating thickness. Plating thickness shall be determined by equation (23).

$$t = t_1' + t_1 + t_2 \tag{23}$$

where,

t = Thickness of outside control surface plating (not to be less than 0.5 inch).

t_1' = Assumed base thickness of outside plating (t_1' may be chosen arbitrarily).

t_1 = 0.06 inch (corrosion allowance)

t_2 = 0.0 inch (cavitation allowance)

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5.6.1.3 Ratio of panel breadth to thickness: breadth b, and length a.

The ratio b/t shall be assumed within the following limits:

$40 \leq b/t \leq 50$, for surface ships only.

If b/t is not within this interval, a new t′ may be assumed, to make b/t equal to the nearer limit.

Length a should be approximately equal to breadth b.

5.6.2 Filled control surfaces. (See 5.6.3.)

5.6.2.1 Ratio of panel breadth to panel thickness, b/t. The ratio b/t shall be chosen within the following limits:

$70 \leq b/t \leq 85$ (for submarines only)

where,

t = Thickness of control surface side plating (to be not less than 0.25 inch)

b = Breadth of panel (inches)

5.6.2.2 Plating thickness, t. Plating thickness for filled control surfaces shall be in accordance with equation (24).

$$t = t' + t_1 + t_2 \tag{24}$$

where,

t′ = 0.19 inch (thickness of plating based on loading)

t_1 = 0.06 inch (corrosion allowance)

t_2 = 0.0 inch (cavitation allowance)

5.6.2.3 Panel breadth and panel length. The length a and breadth b should be approximately equal.

5.6.3 Other considerations (see figure 15). Additional considerations that shall be taken into account in determining web spacing and web and plate thickness are specified in 5.6.3.1 through 5.6.3.4.

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5.6.3.1 Trailing edge. At the trailing edge of the control surface a rabbeted casting or forging shall be used.

5.6.3.2 Leading edge. The leading edge of the control surface shall be rolled plate of at least the maximum thickness of the outside plating.

5.6.3.3 Web thickness.

a. Web thickness in nonfilled control surfaces shall be not less than 70 percent of the side plating thickness.

b. Web thickness in filled control surfaces shall be not less than 0.25 inch.

5.6.3.4 Variation in web spacing. Web spacings determined by the foregoing methods may require modification in way of hub castings.

5.7 Determination of hub casting thickness. A hub casting shall be incorporated into the structure of the control surface (for exception, see 5.7.3). Functions of the casting are to provide means of accepting the stock, transmitting torque between the stock and control surface, helping to resist the bending moment, and attaching the control surface to the stock. An example of a hub casting and its attachment to webs and plating is shown on figure 15 for a spade rudder.

5.7.1 General characteristics. An appropriate hub casting to perform the above-cited functions shall have the following characteristics:

a. The stock shall penetrate the hub no less than two stock diameters as measured at the root chord.

b. The minimum hub wall thickness shall be not less than 0.15 stock diameter.

c. The taper of the hub-to-stock interface shall be between 1.5 and 2.0 inches (inclusive) per foot of stock diameter.

d. If relief is provided along the hub-to-stock interface, no more than the middle third of the hub shall be relieved.

5.7.2 **Local stress.** Local stress calculations of the hub need not be performed except where keys and keyways are located or where some unconventional arrangement is used; however, hub strength shall be considered in determining overall bending and shear of the control surface. The moment of inertia of the hub and associated stock used in these calculations shall be as given by equation (25).

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$$I_H = \frac{I_{H\text{-}A}}{K} \tag{25}$$

where,

$I_{H\text{-}A}$ = Actual moment of inertia of the hub and associated stock about centroidal axis of control surface section (in^4).

K = Stress factor depending on stock penetration (see figure 16).

I_H = Moment of inertia of hub and associated stock to be used in strength calculations in^4.

5.7.3 Dispensing with hub casting. If shipping and unshipping of the rudder in a drydock does not require special high blocking, the hub casting may be eliminated. In that event the rudder stock shall be welded onto the rudder horizontal webs as shown on the latter part of figure 15. In such cases, the rudder stock shall have machined lands or bosses onto which the rudder horizontal webs are welded. At least four horizontal webs shall be welded onto the rudder stock, and the penetration of the stock into the rudder shall be at least 2.5 times the rudder stock diameter at the top of the rudder. The thickness of the webs attached to the rudder stock shall be 50 percent thicker than the webs used with a hub casting; the total spanwise length of these thicker webs shall be not less than 2.0 times the rudder stock diameter at the top of the rudder (similarly for other types of control surfaces).

5.8 Spanwise and chordwise stresses on control surface body. The spanwise and chordwise stresses shall be determined by using the moments and shears from 5.1. Stresses shall be calculated for suspected critical areas, especially where a sudden change in moment of inertia takes place.

5.9 Gudgeons, pintles, keyways, grooves, and discontinuities. These stress regions shall be treated as specified in 5.9.1 through 5.9.4.

5.9.1 Gudgeons. The stress in a gudgeon may be determined from equation (26).

$$\sigma_G = \frac{GT}{10aL} \tag{26}$$

where,

$$a = \frac{R_2 + R_1}{20} \qquad (inches)$$

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R_1 = Outside radius of gudgeon (inches)

R_2 = Inside radius of gudgeon (inches)

L = Length of gudgeon (inches)

T = Total reaction at gudgeon (or shear force at horn) (kip)

G = Gudgeon strength factor (see figure 17)

σ_G = Maximum stress in gudgeon (kip/in^2) (see 4.2.5)

5.9.2 Pintles. In the design of pintles only the shear stresses in the pintle need be considered. A factor of safety of 5.0 or greater on yield shall be provided.

5.9.3 Keyways. Where keys are used with stocks to transmit torque between the stock and the control surface, stress concentration in the stock due to the keyway shall be allowed for.

5.9.3.1 Stress concentration. A stress concentration factor of 1.0 may be assumed for bending. Torsional stress concentration factors can be determined from figure 18. Torques and bending moments in way of keyways shall be multiplied by their respective factors in order to determine effective stresses.

5.9.4 Grooves and discontinuities. For grooves and discontinuities such as those for rudder carrier plates, stress concentration factors shall be applied (see figure 19). The location of a groove on the rudderstock could be of critical importance because it is a high stress riser. If the situation were such that the location of the groove caused high stress, a built-in flange for the carrier plate instead of a groove could be used, but shipping and unshipping of the stock would need to be reanalyzed to ensure ease of maintainability.

5.10 Cathodic protection. Corrosion protection of control surfaces is provided by galvanic anode protectors (zincs). Forty percent of the hull anodes shall be installed in the stern area in a location not interfering with the struts, propellers, propeller shafts, or rudders. Anodes shall be installed equally divided, port and starboard, on the hull adjacent to the propellers in locations always submerged when the ship is afloat but not making way. For submarines zinc anodes shall be attached to the hull plating in the stern area, approximately 5 feet forward of the propeller planes. Zinc anodes shall not be attached to rudders or diving planes.

5.11 Determination of rudder scantlings by ABS rules. For auxiliary ships that are required to meet the requirements of the American Bureau of Shipping the rudder scantling shall meet at least the requirements of ABS Rules for Building and Classing Steel Vessels or other ABS publications for special ship types.

5.12 Strengthening for navigation in ice. Ice-strengthening shall depend on whether the ship is an auxiliary or a surface combatant, as specified in 5.12.1 or 5.12.2.

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5.12.1 Auxiliary ships. If an auxiliary ship is to navigate in ice and to receive special classification for such service, then the strength of the rudders and steering gear shall be increased over that required by ABS rules for Building and Classing Steel Vessels. ABS rules specify increased strength of the rudder stock and thickness of rudder plating depending on the class of ice-strengthening required.

5.12.2 Surface combatant ships. Navy surface combatant ships do not normally require special strengthening for navigation in ice. If ice-strengthening is required the guidelines provided in ABS Rules for Building and Classing Steel Vessels may be used.

5.12.3 Submarines. Ice loads are considered for the fairwater planes of some submarines which must surface through ice. The planes are made rotatable with the leading edge turning up. Typical ice breathrough loads are determined for the rotated fairwater planes on an individual ship basis.

5.13 Examples. Numerical examples of design calculations are presented in appendixes as follows:

Appendix A. Surface ship spade rudder.

Appendix B. Surface ship horn rudder.

Appendix C. Submarine stern diving plane.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended for use in designing control surface structures for naval ships, including submarines.

6.2 Issue of DODISS. When this standard is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1 and 2.3).

6.3 Additional references. The documents cited below provide general background on the topics of this military standard.

Lewis, Edward V., editor, *Principles of Naval Architecture*, The Society of Naval Architects and Marine Engineers.

Taggart, Robert, editor, *Ship Design and Construction*, The Society of Naval Architects and Marine Engineers.

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6.4 Subject term (key word) listing.

Fin stabilizers
Diving planes
Rudders

Preparing activity:
Navy–(SH)
(Project 1990-N088)

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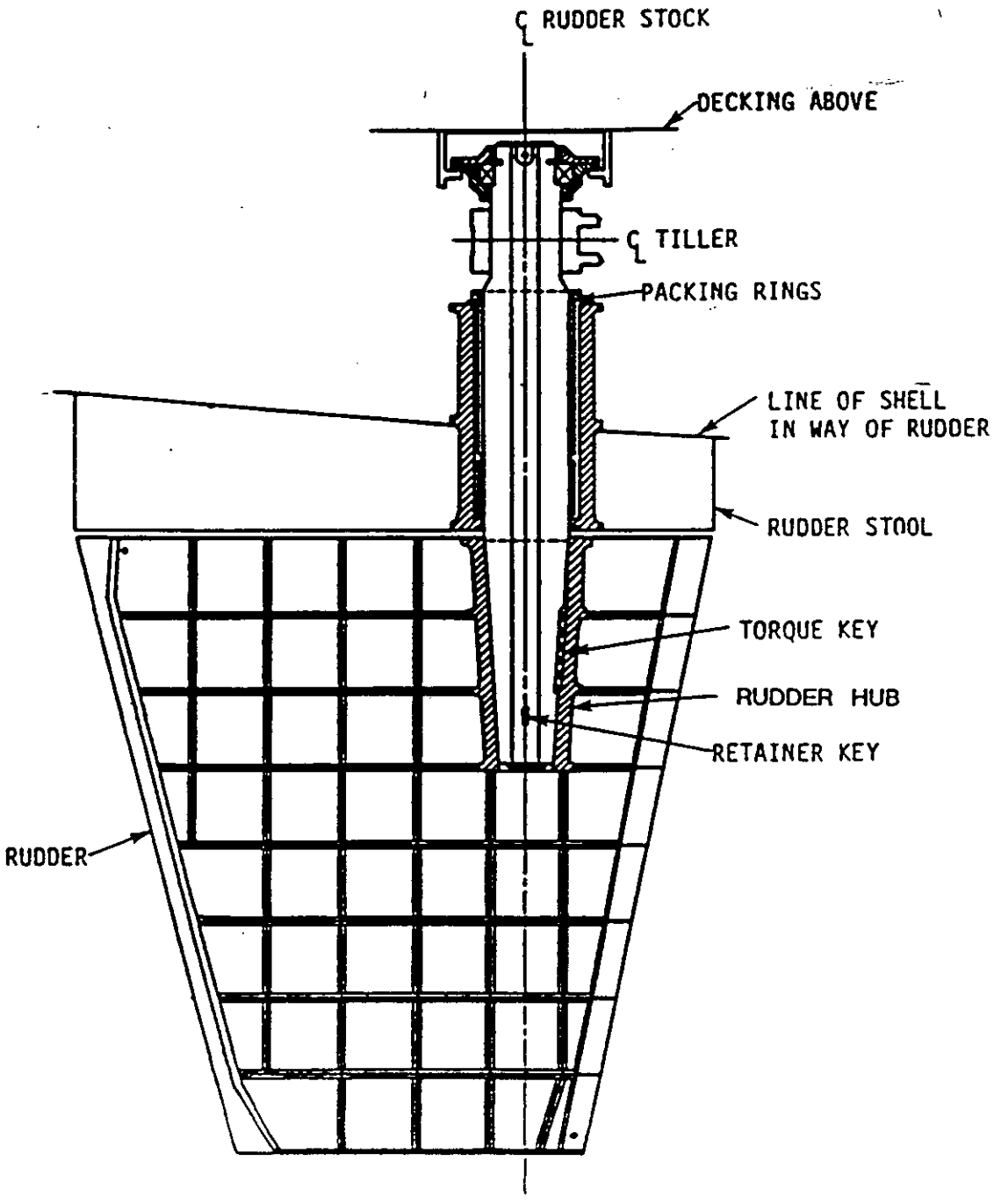


FIGURE 1. *Spade rudder.*

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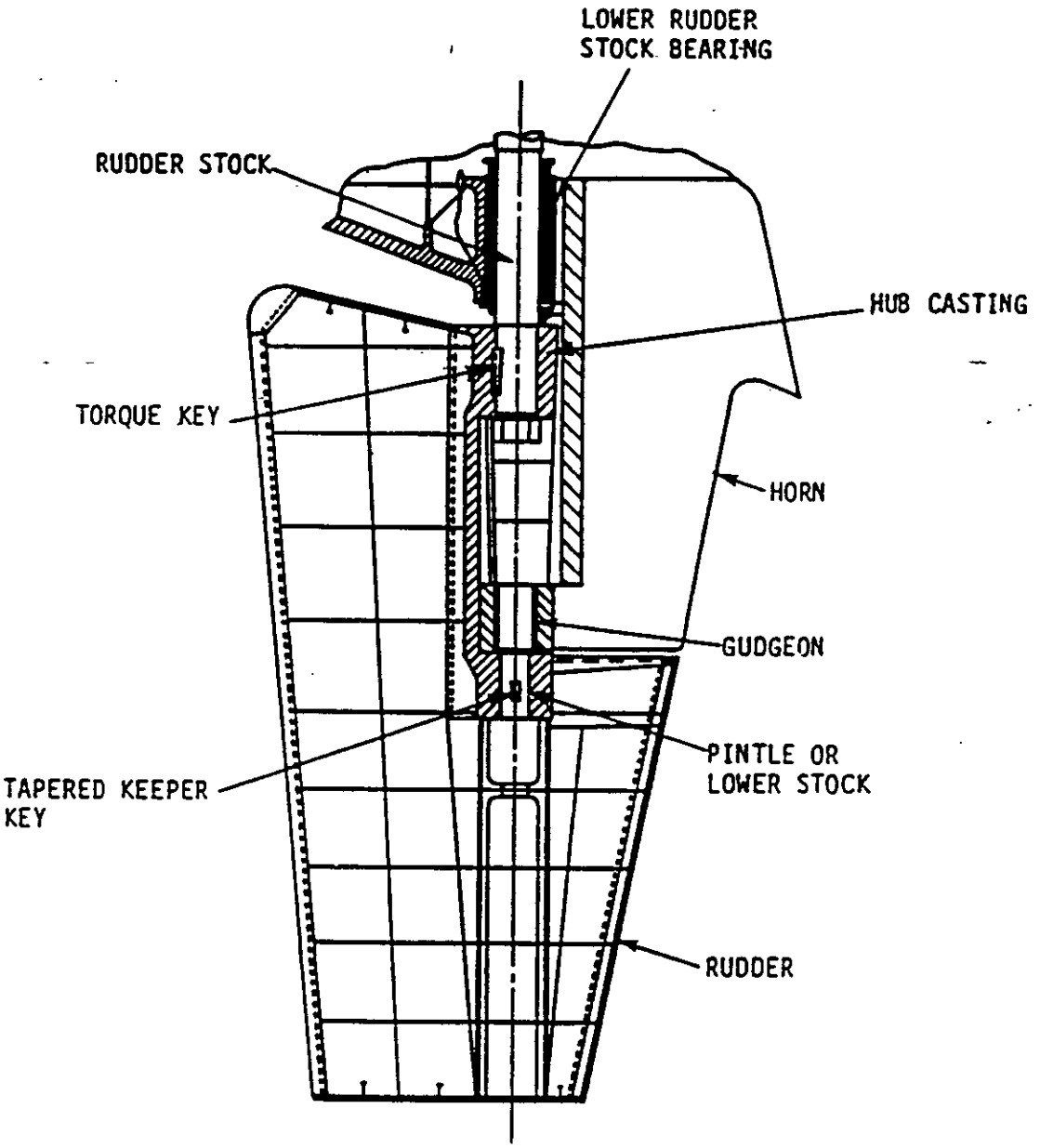


FIGURE 2. *Horn rudder.*

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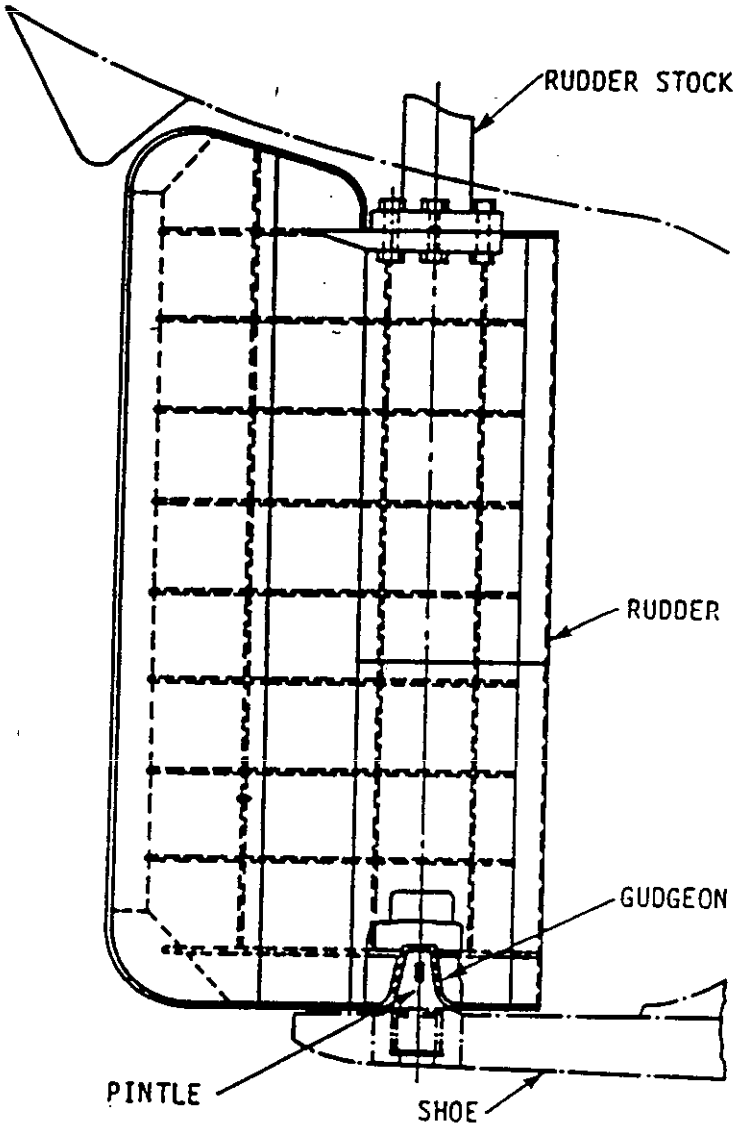


FIGURE 3. *Balanced rudder.*

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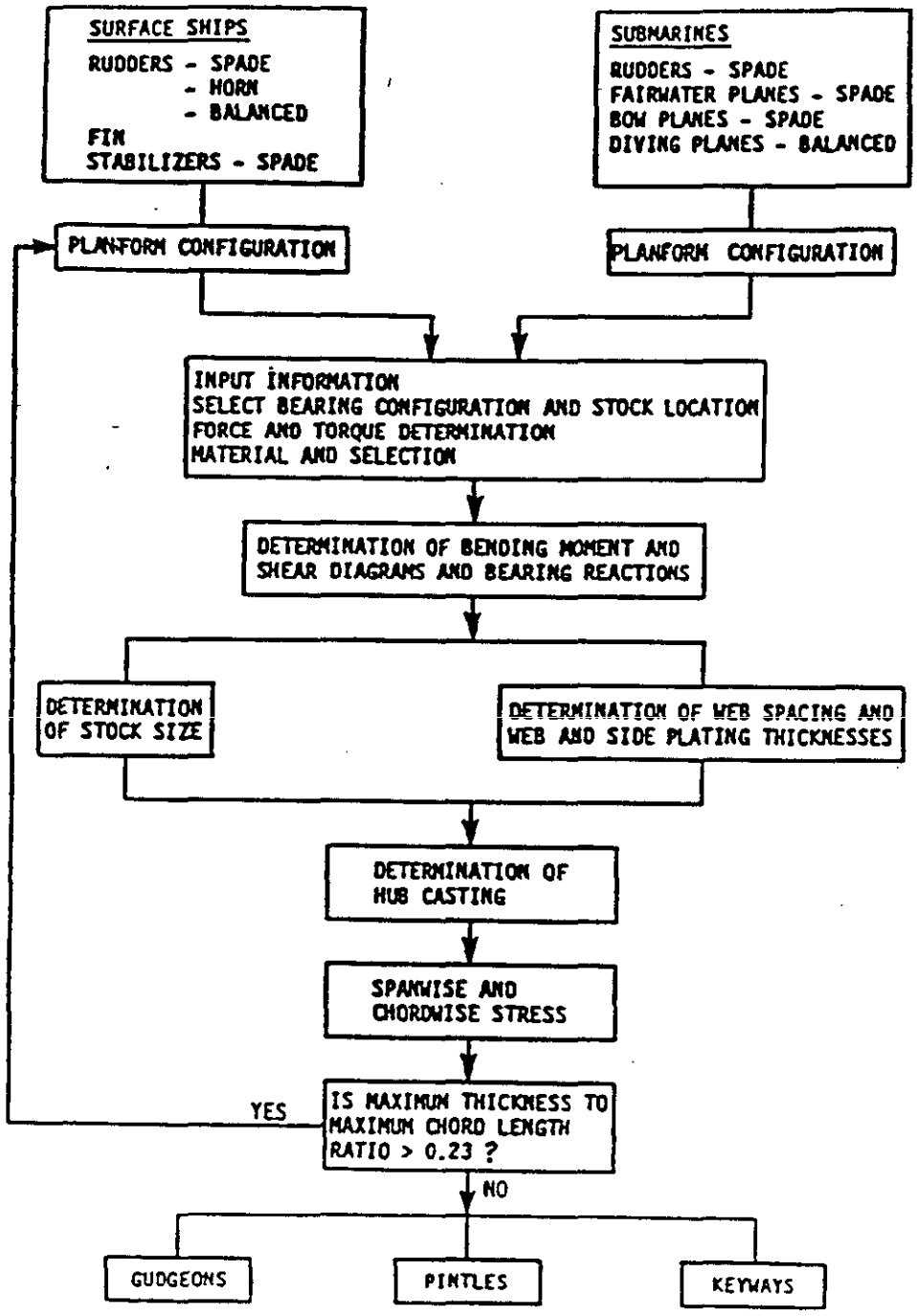


FIGURE 4. *Design sequence flow diagram.*

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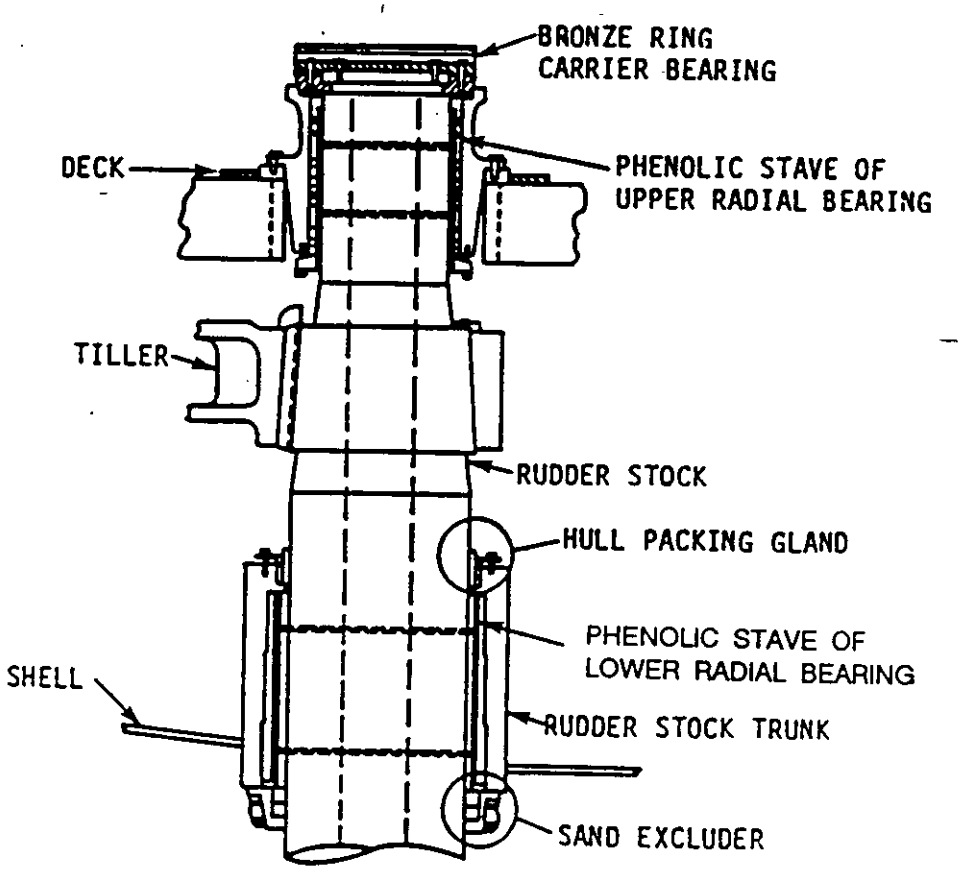


FIGURE 5. *Rudder stock bearing, typical arrangement A.*

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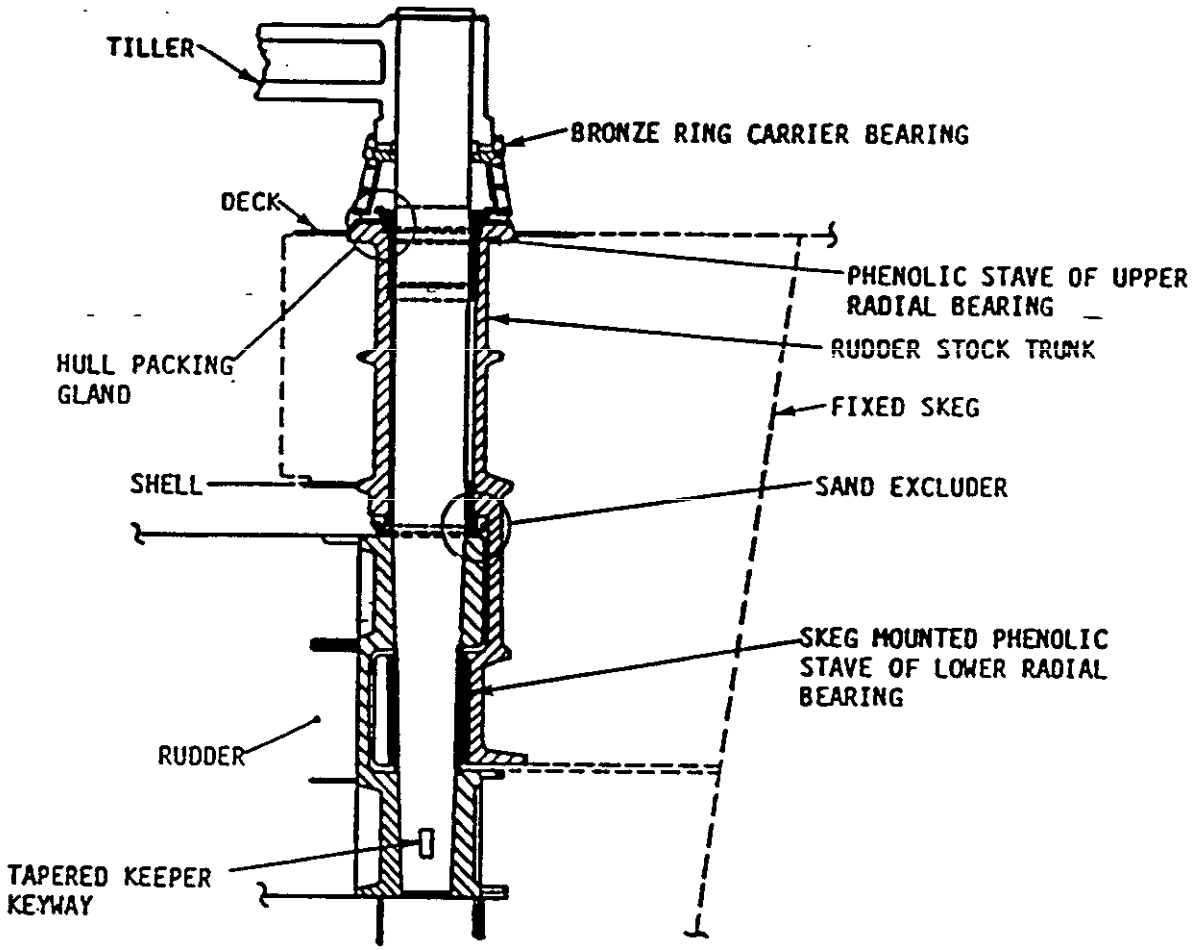


FIGURE 6. *Rudder stock bearing, typical arrangement B.*

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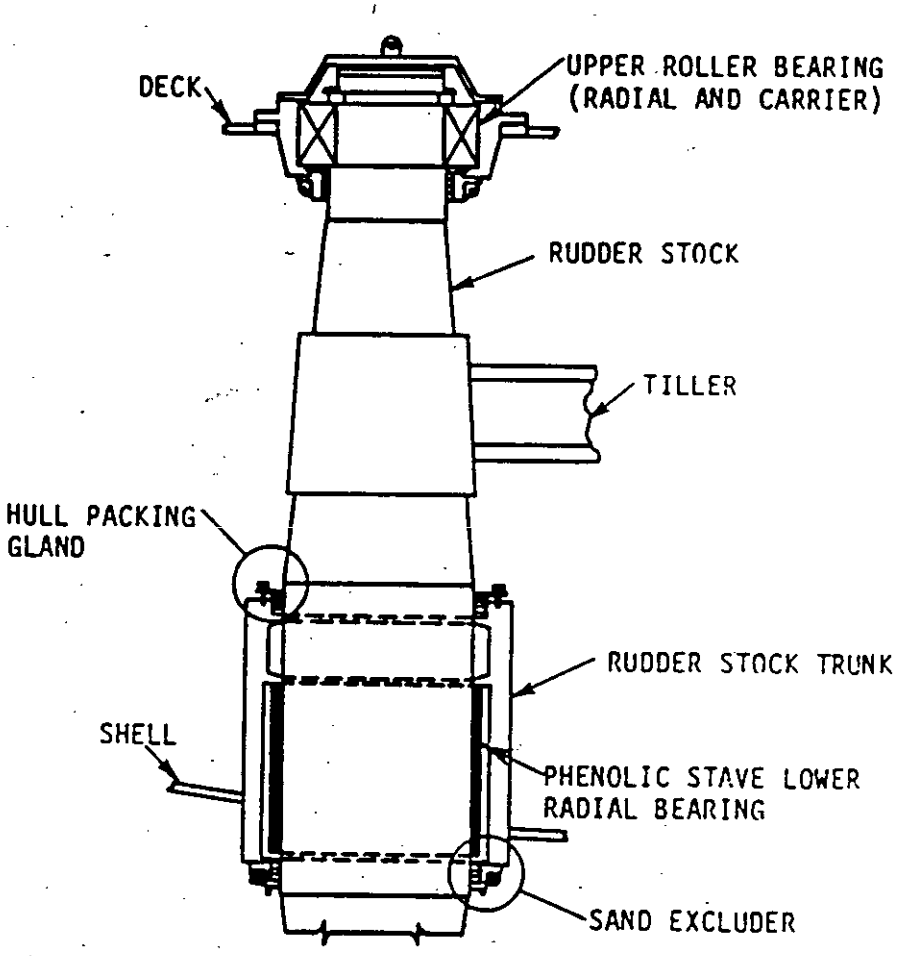


FIGURE 7. *Rudder stock bearing, typical arrangement C.*

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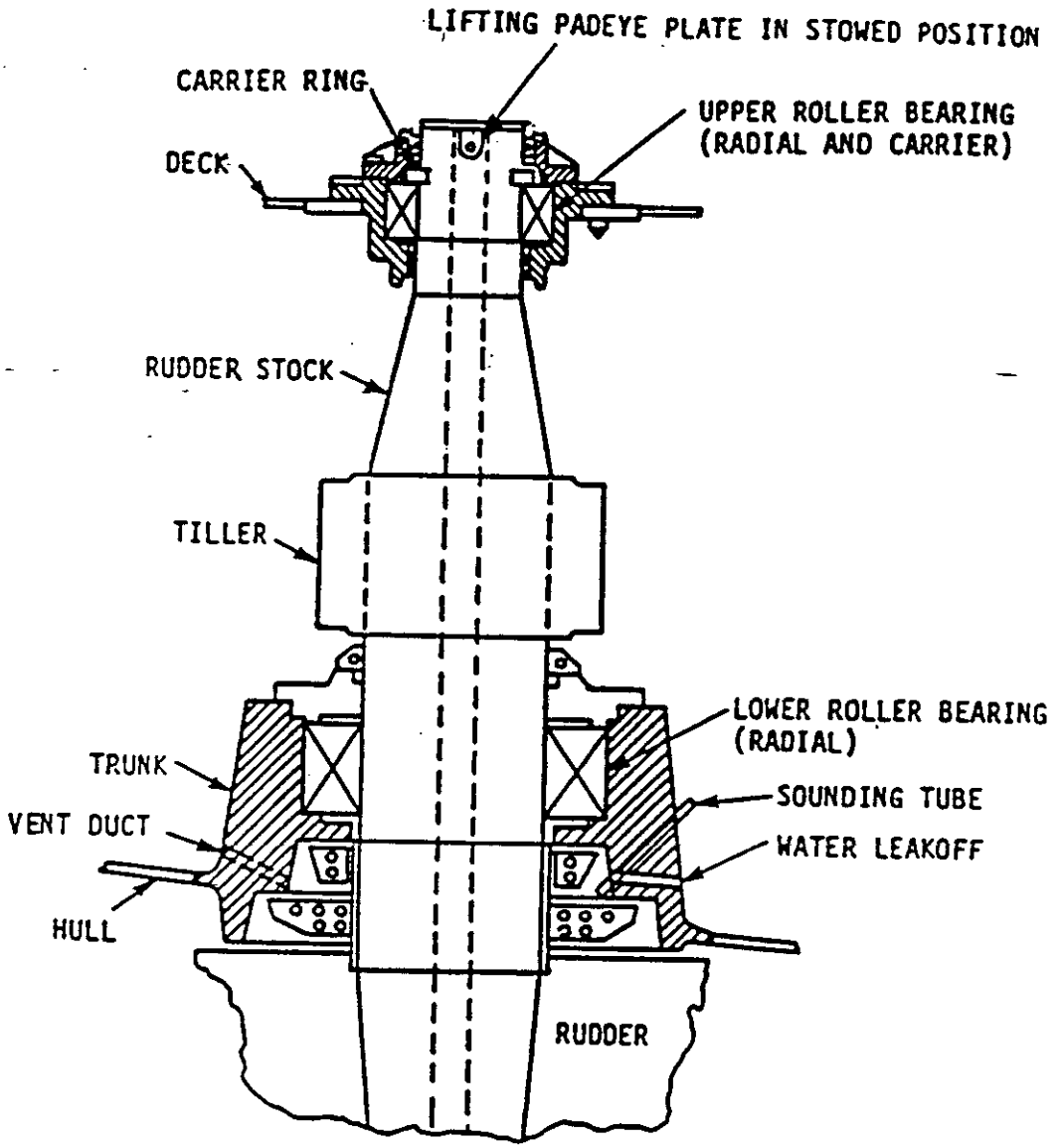


FIGURE 8. *Rudder stock bearing, typical arrangement D.*

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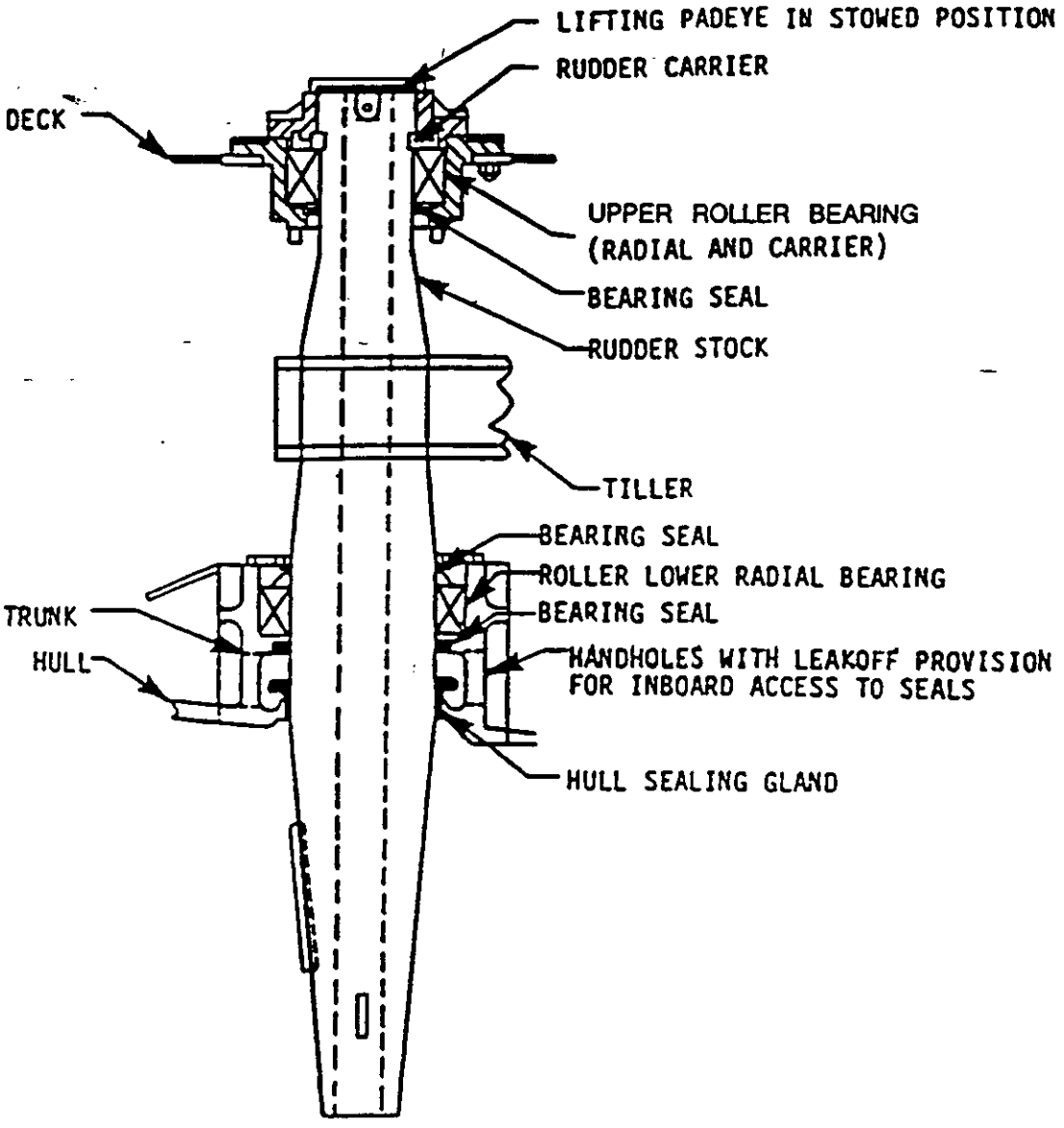


FIGURE 9. *Rudder stock bearing, typical arrangement E.*

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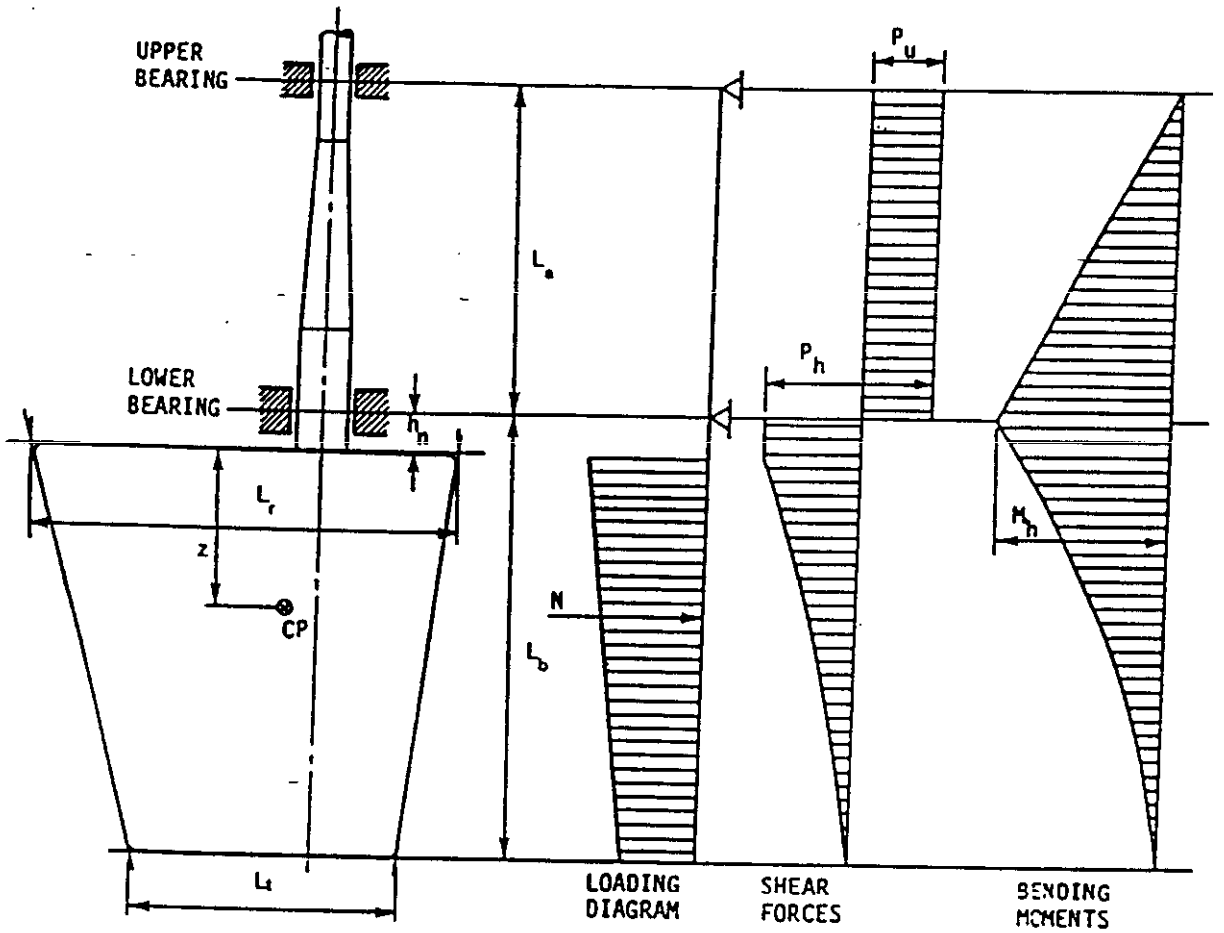


FIGURE 10. *Spade-type control surface, with spanwise load, shear, and bending-moment diagrams.*

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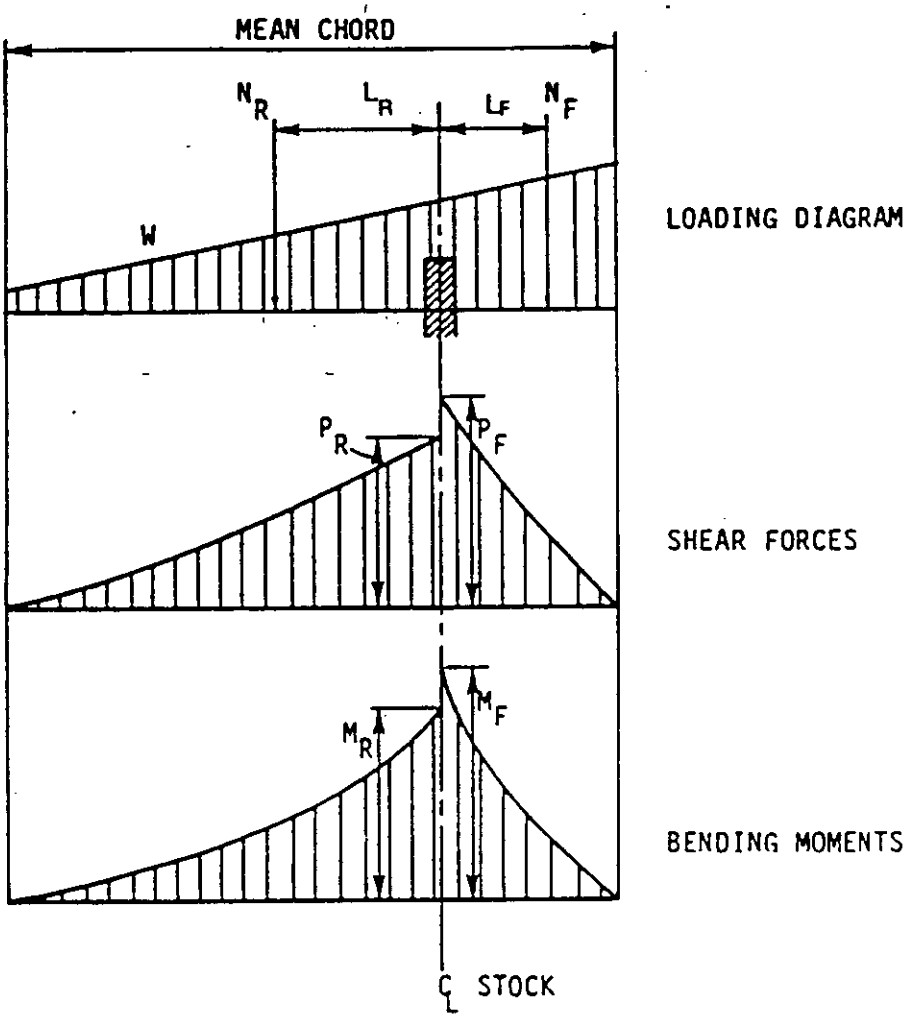


FIGURE 11. *Spade-type control surface, chordwise loading, shear, and bending-moment diagrams.*

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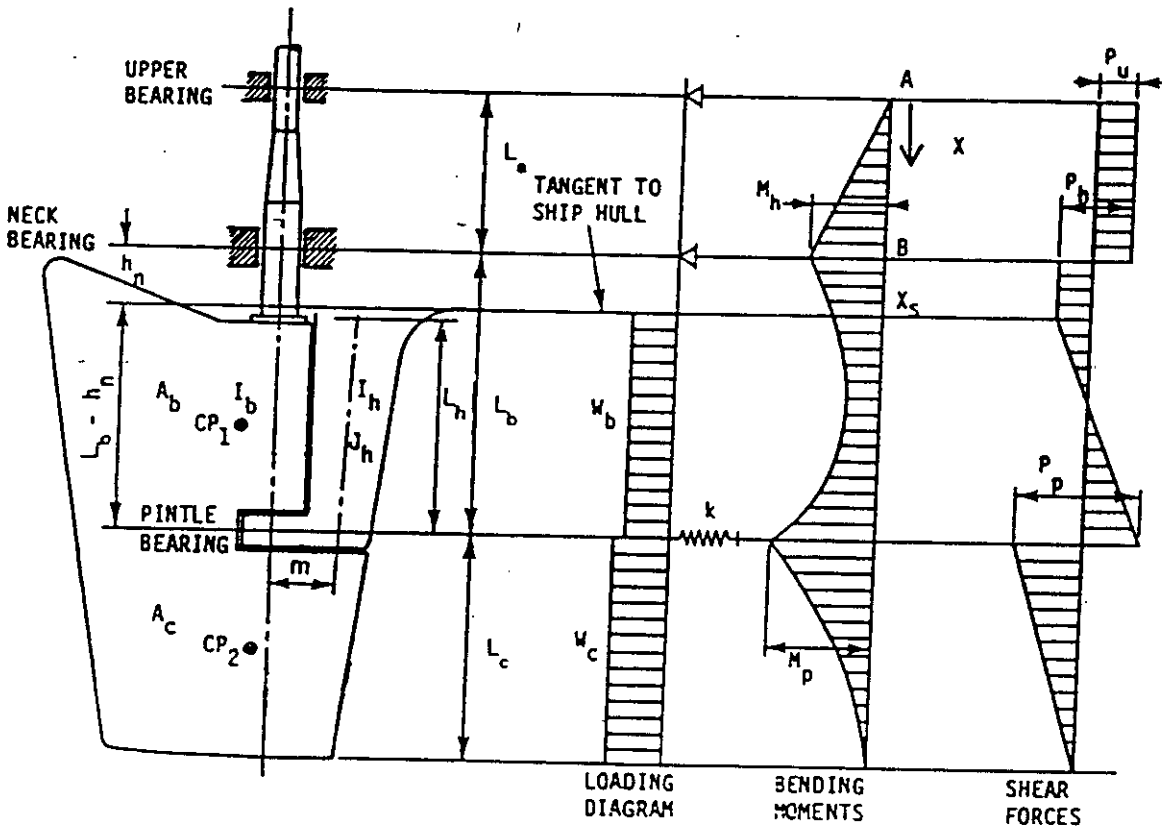
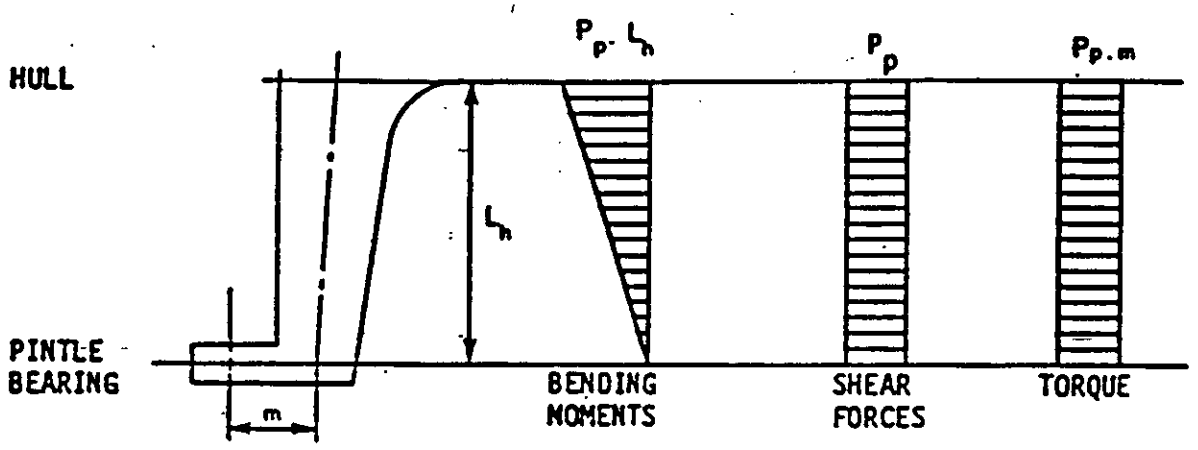


FIGURE 12. *Horn-type control surface, with spanwise loading, bending-moment, and shear diagrams.*

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MAXIMUM BENDING MOMENT $= P_p \cdot L_h$ kip - ft

SHEAR FORCE $= P_p$ kip

TORQUE $= P_p \cdot m$ kip - ft

FIGURE 13. *Control-surface horn, with bending-moment, shear, and torque diagrams.*

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Check the source to verify that this is the current version before use.

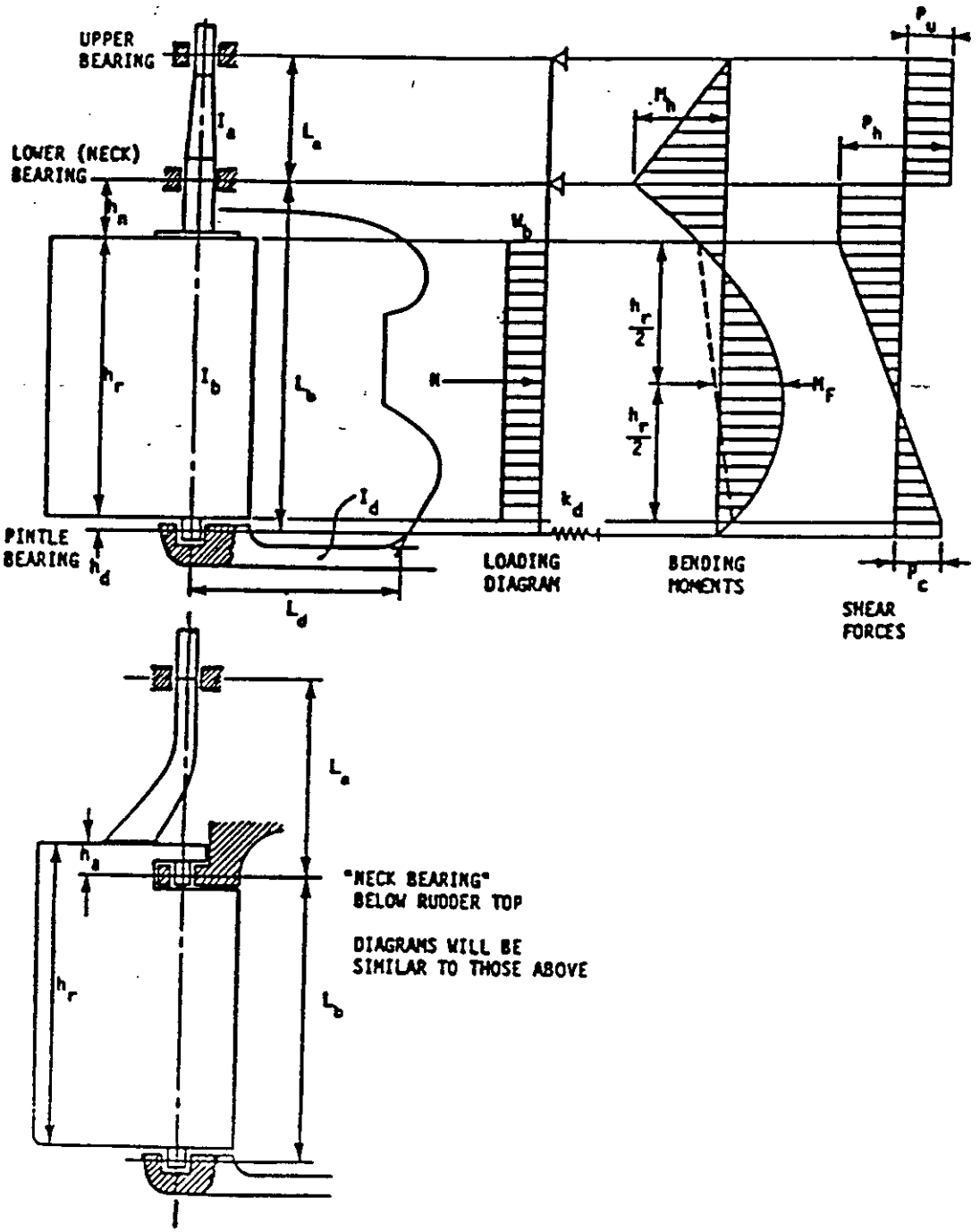


FIGURE 14. *Balanced-type control surfaces, with spanwise loading, bending-moment, and shear diagrams.*

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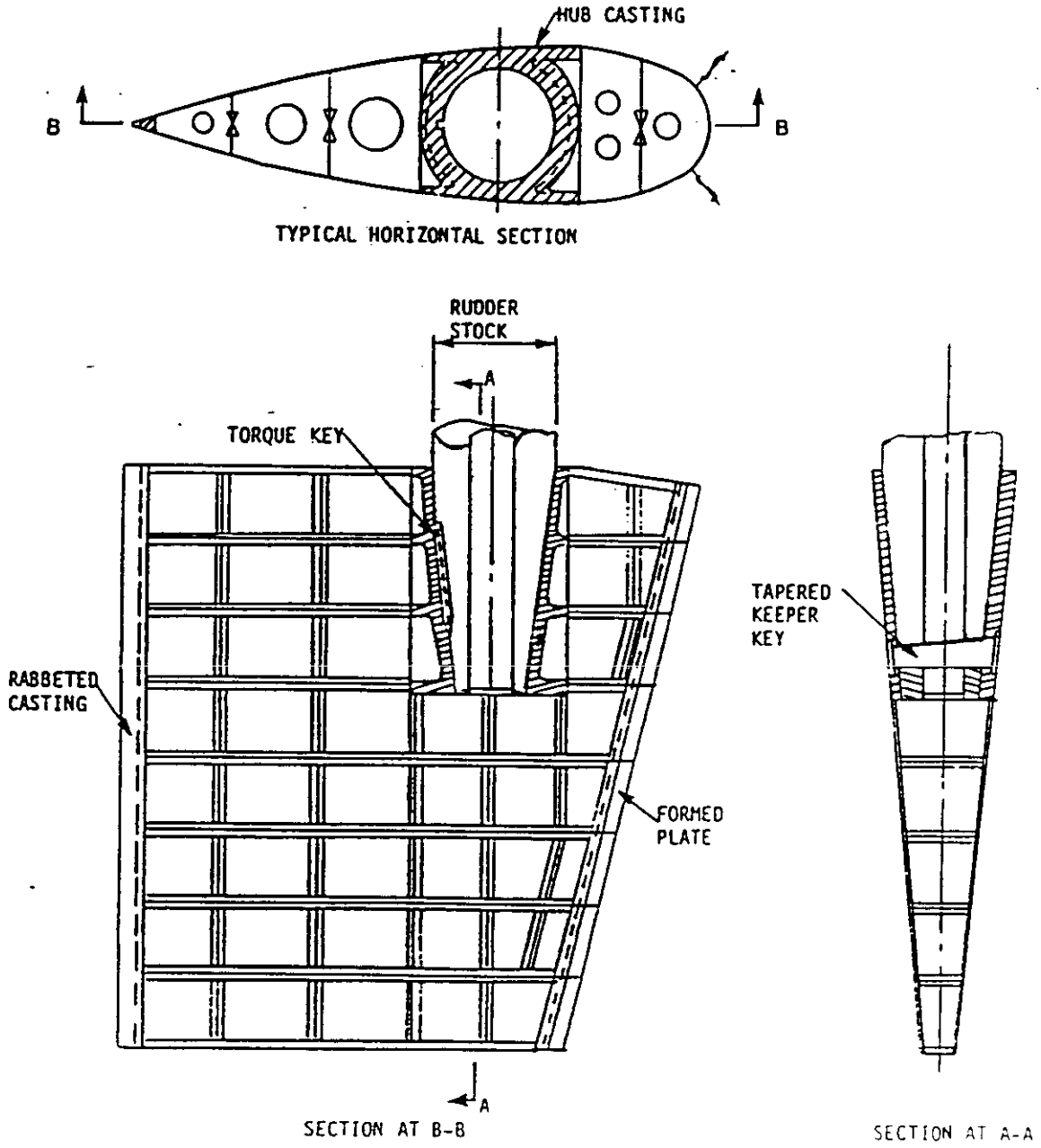


FIGURE 15. *Hub casting and integral stock.*

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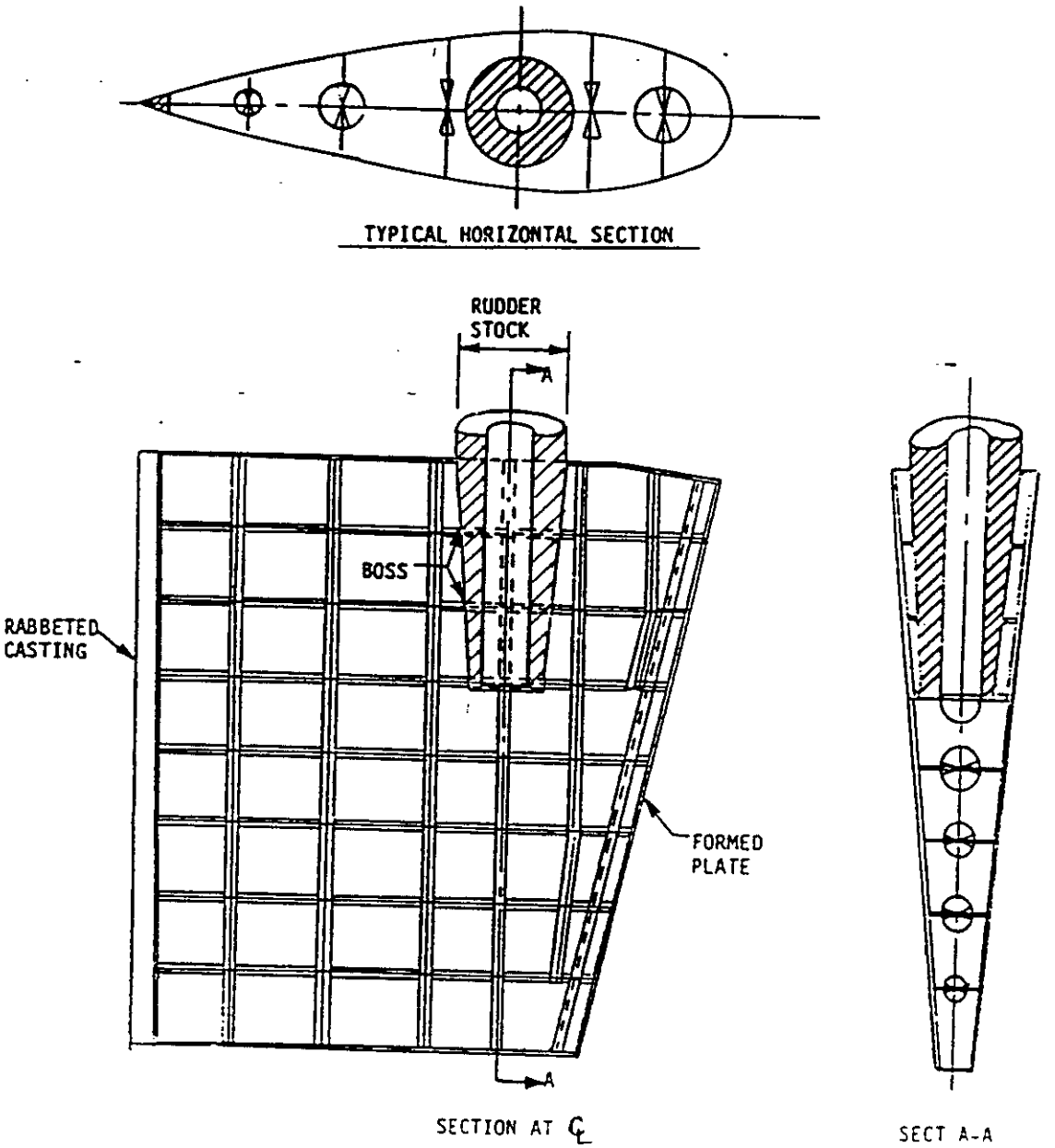


FIGURE 15. *Hub casting and integral stock. – Continued*

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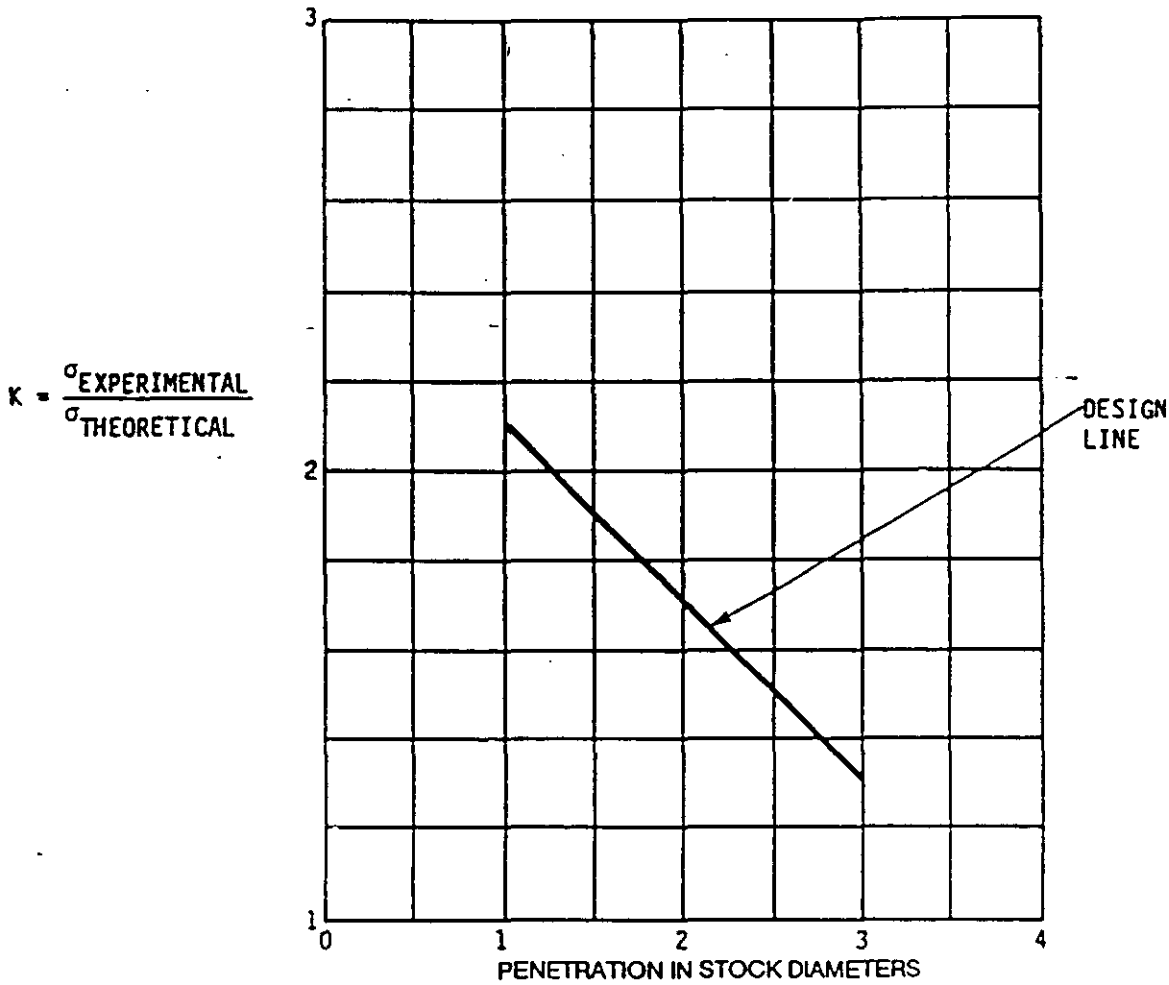


FIGURE 16. *Stress factor K plotted as function of penetration.*

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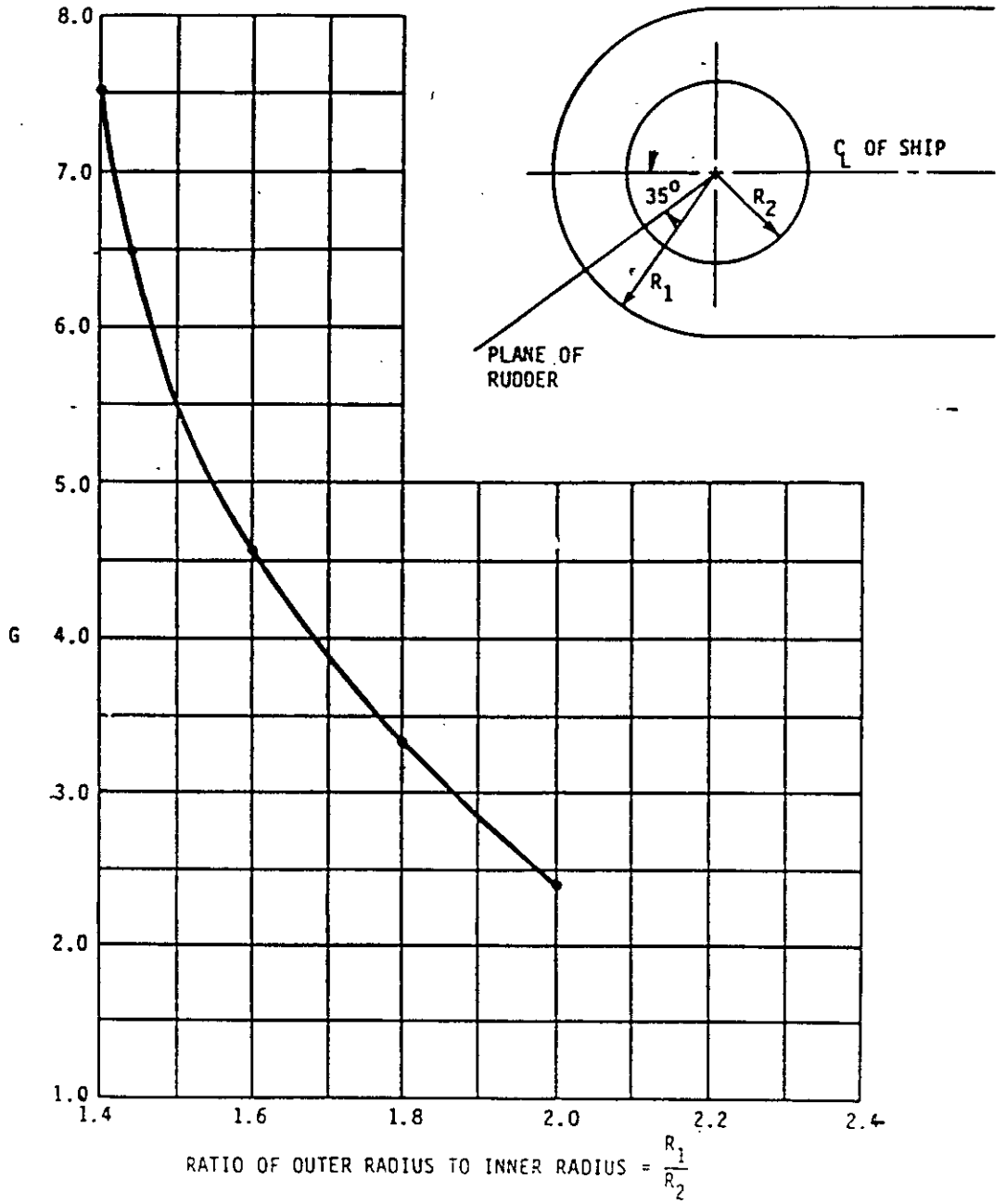


FIGURE 17. *Gudgeon strength factor G plotted as function of ratio of outer to inner radius.*

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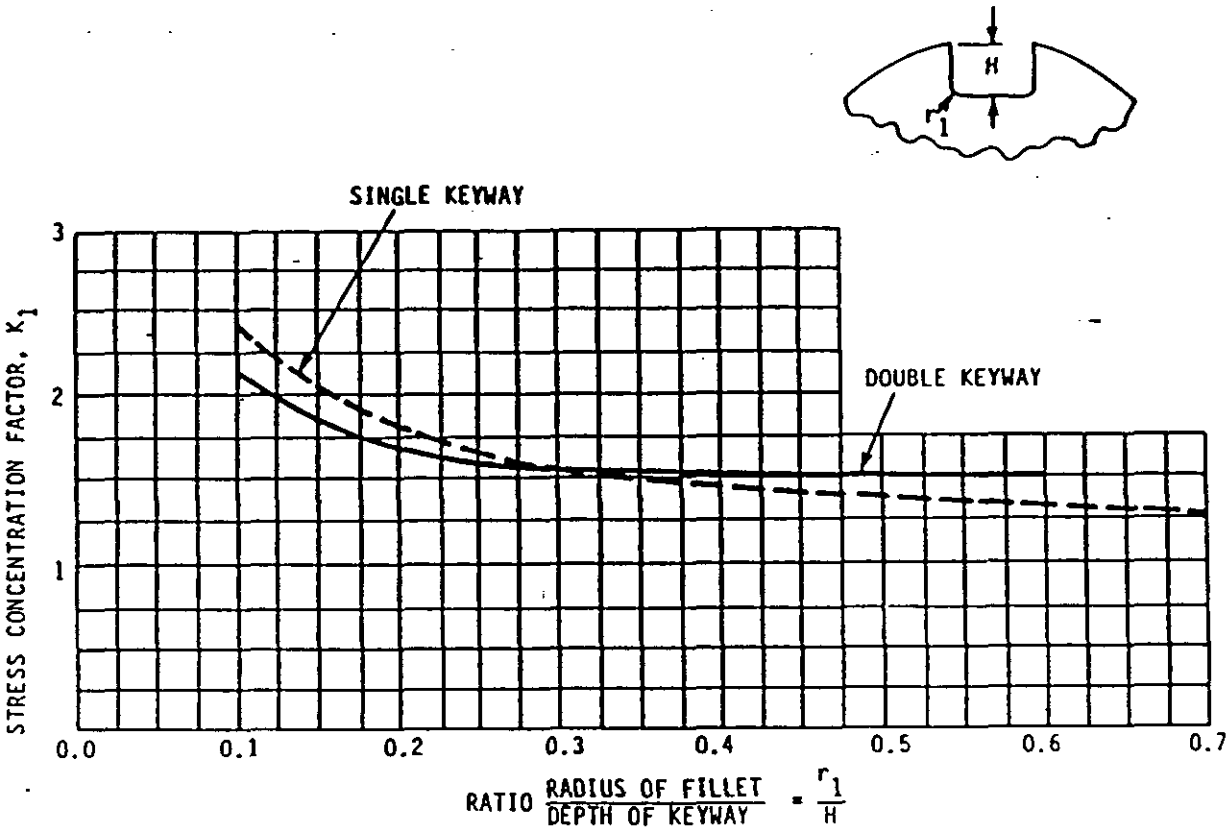


FIGURE 18. *Stress concentration factor at keyway fillet, in torsion.*

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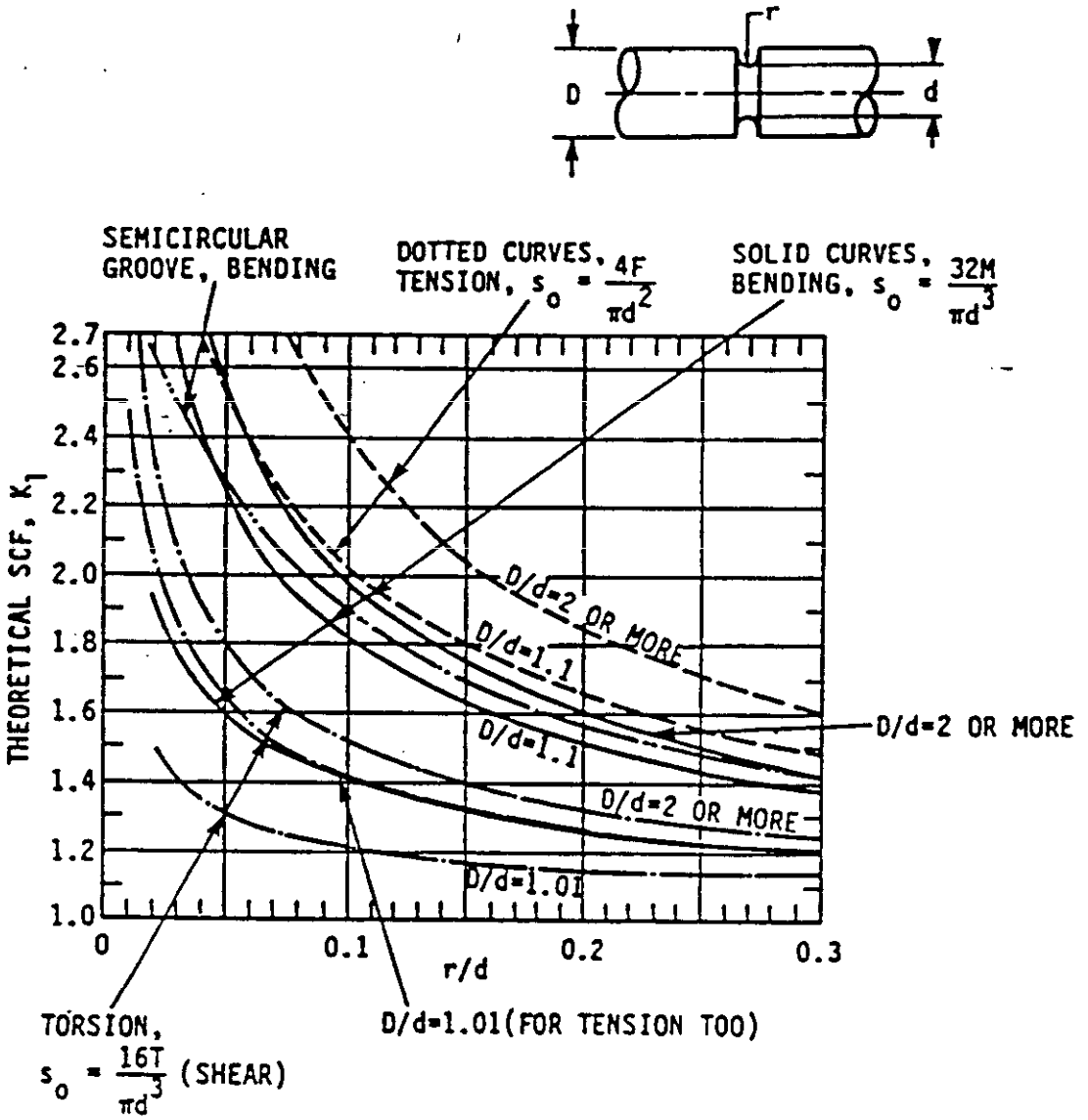


FIGURE 19. *Stock with groove, theoretical stress concentration factor K_1 plotted as function of ratio of fillet radius of groove to reduced diameter of stock.*

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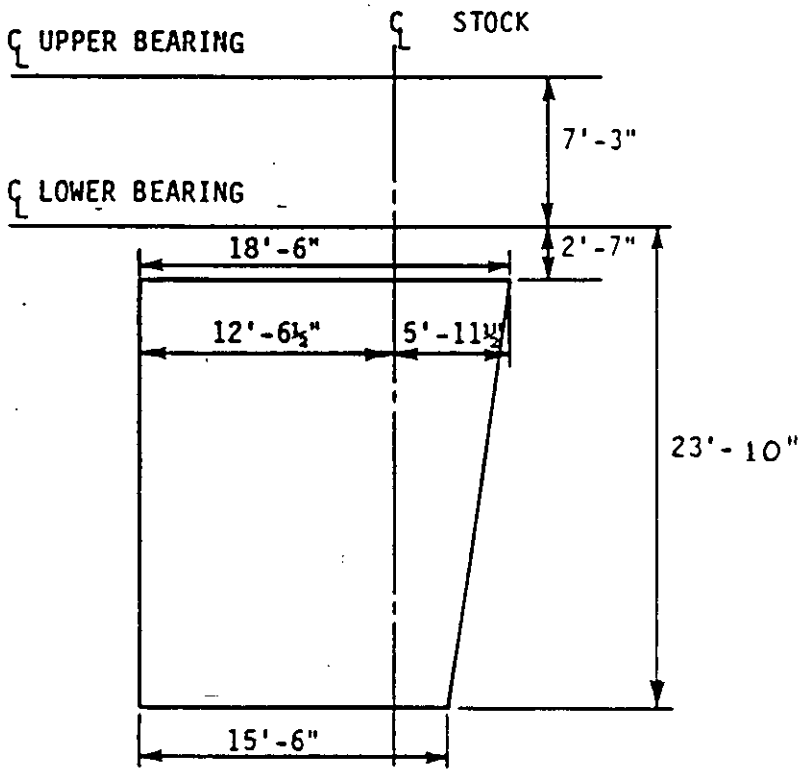


FIGURE 20. *Spade rudder dimensions – example.*

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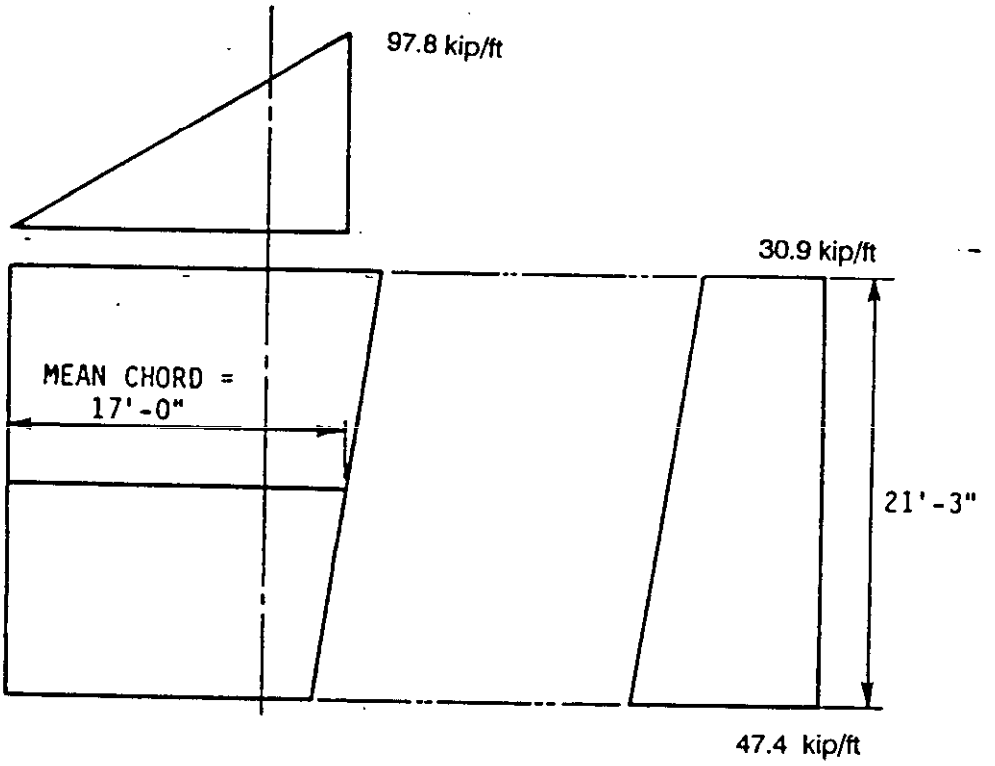


FIGURE 21. *Spade rudder spanwise and chordwise hydrodynamic load distributions – example.*

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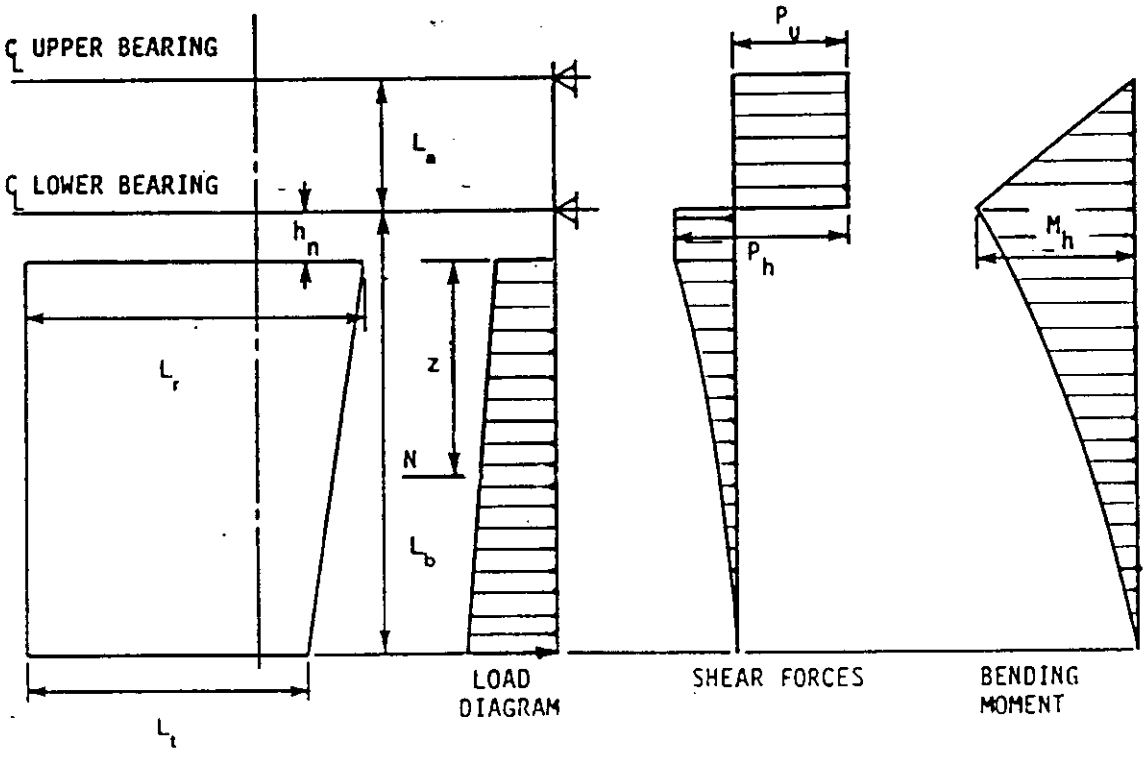


FIGURE 22. *Spade rudder spanwise hydrodynamic loading – example.*

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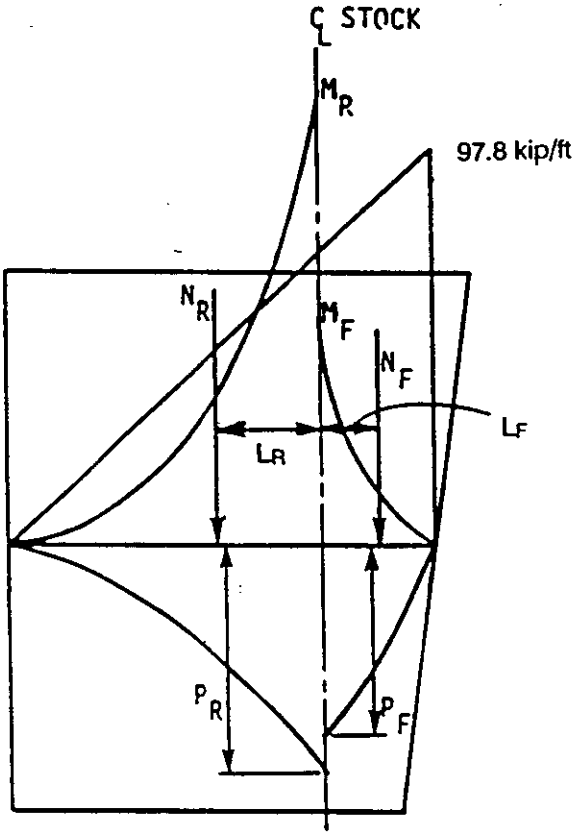


FIGURE 23. *Spade rudder chordwise hydrodynamic loading – example.*

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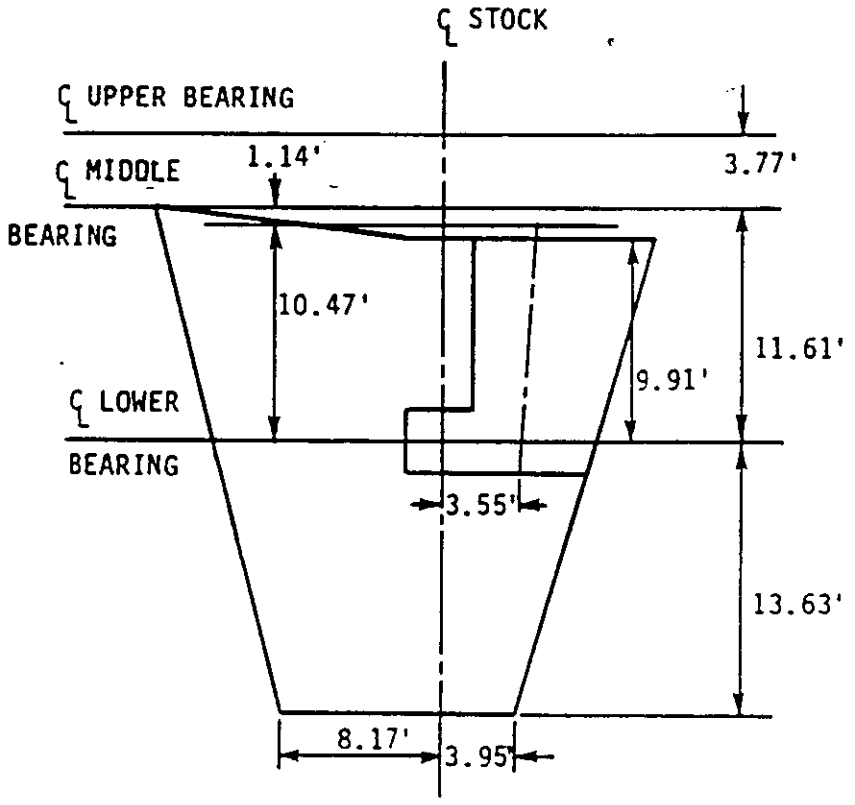


FIGURE 24. *Horn rudder dimensions – example.*

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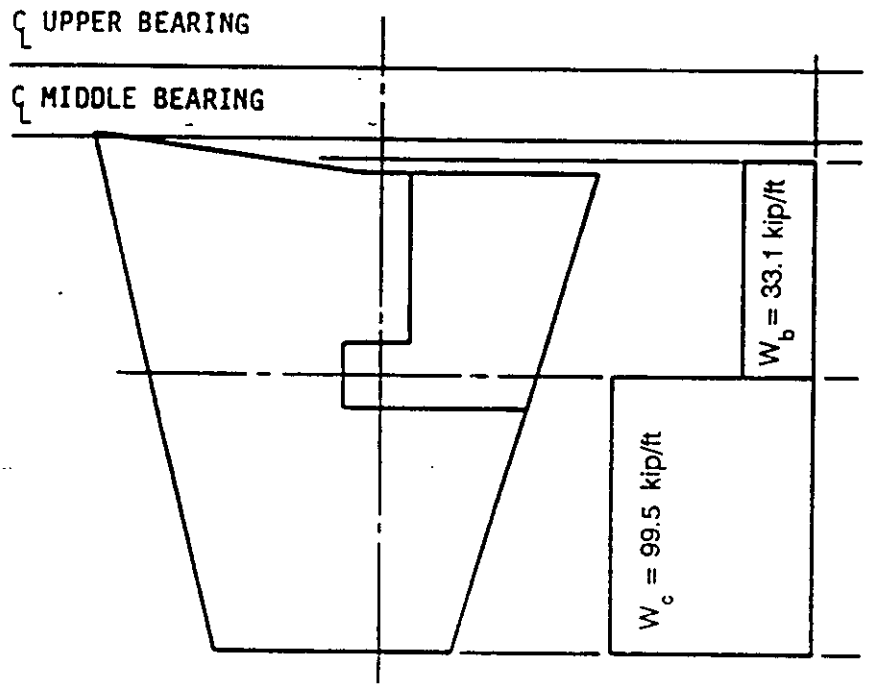


FIGURE 25. *Horn rudder spanwise hydrodynamic loading – example.*

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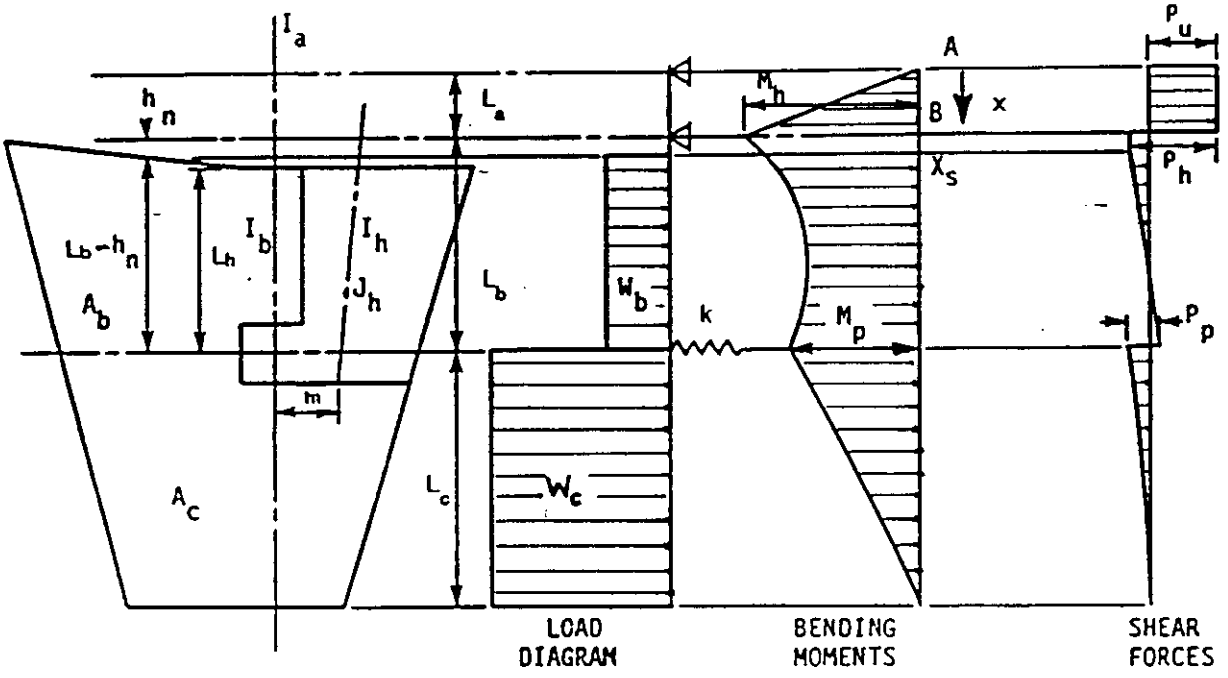


FIGURE 26. *Horn rudder shear and bending moment diagrams and bearing reactions – example.*

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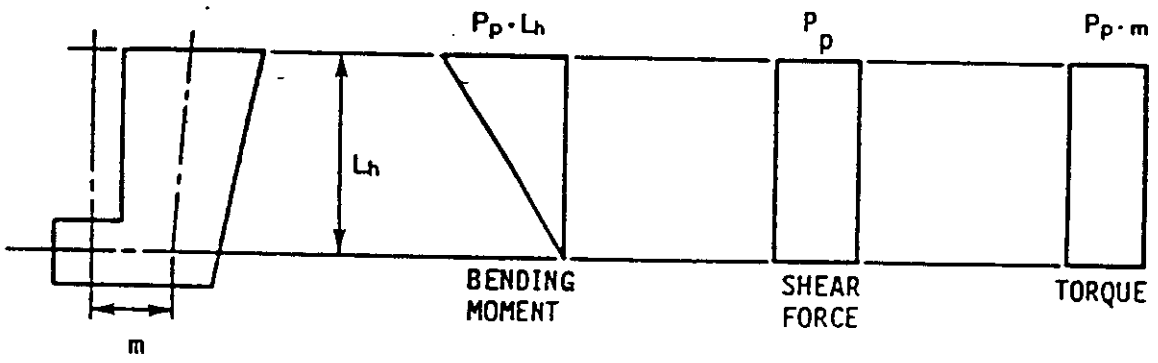


FIGURE 27. *Control surface horn: Bending moment, shear force, and torque distributions – example.*

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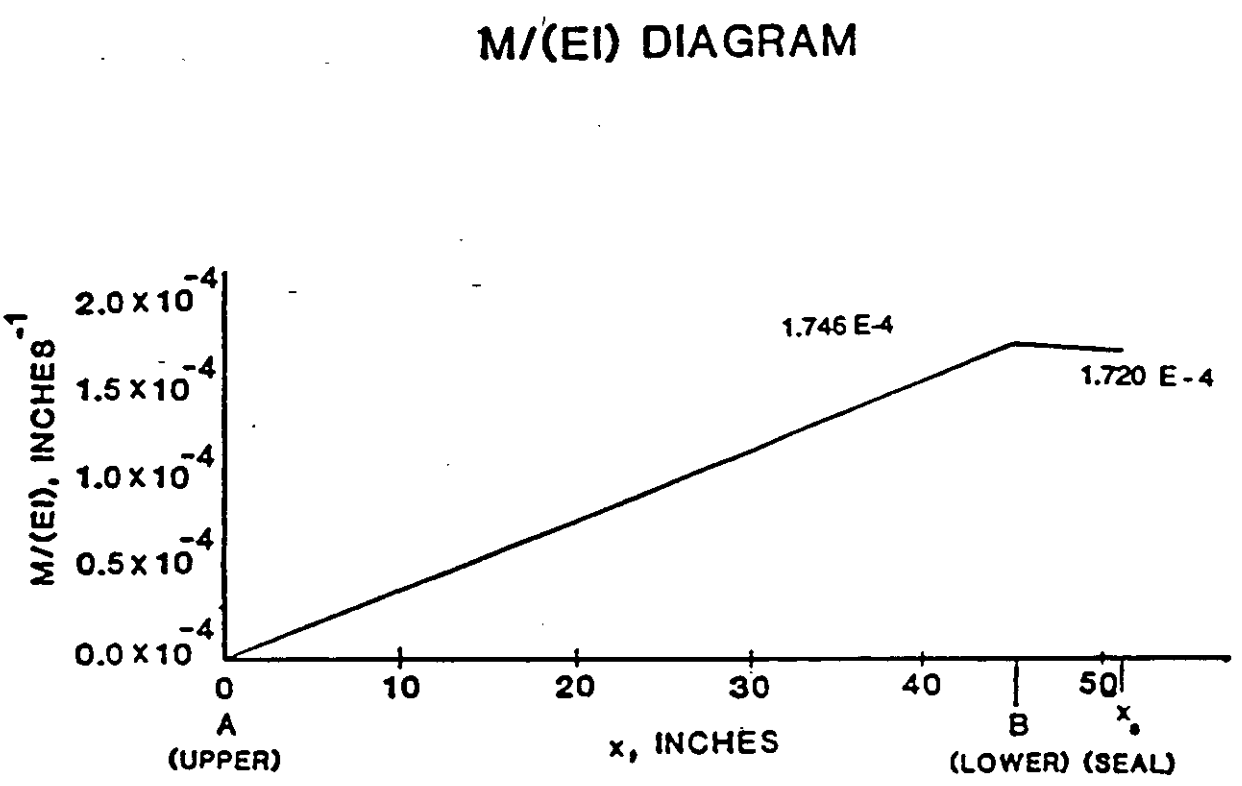


FIGURE 28. *M/EI diagram – example.*

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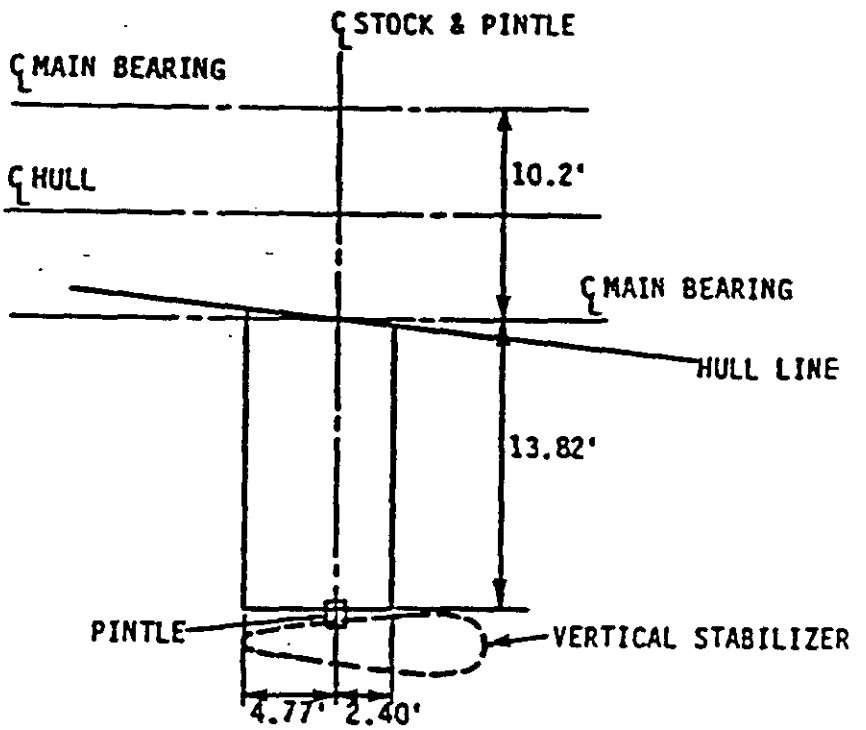


FIGURE 29. *Balanced diving plane dimensions – example.*

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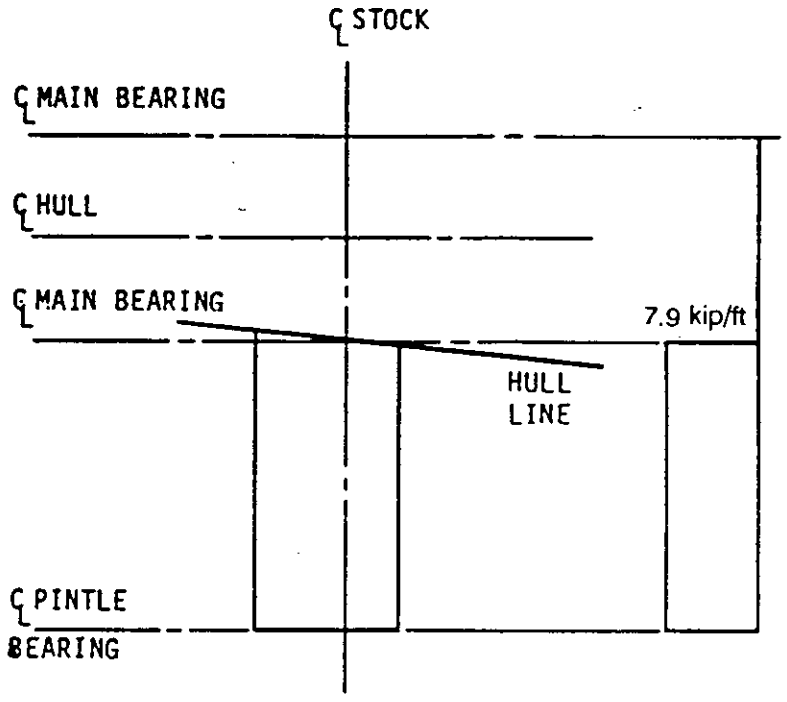


FIGURE 30. *Balanced diving plane spanwise sea slap loading distribution – example.*

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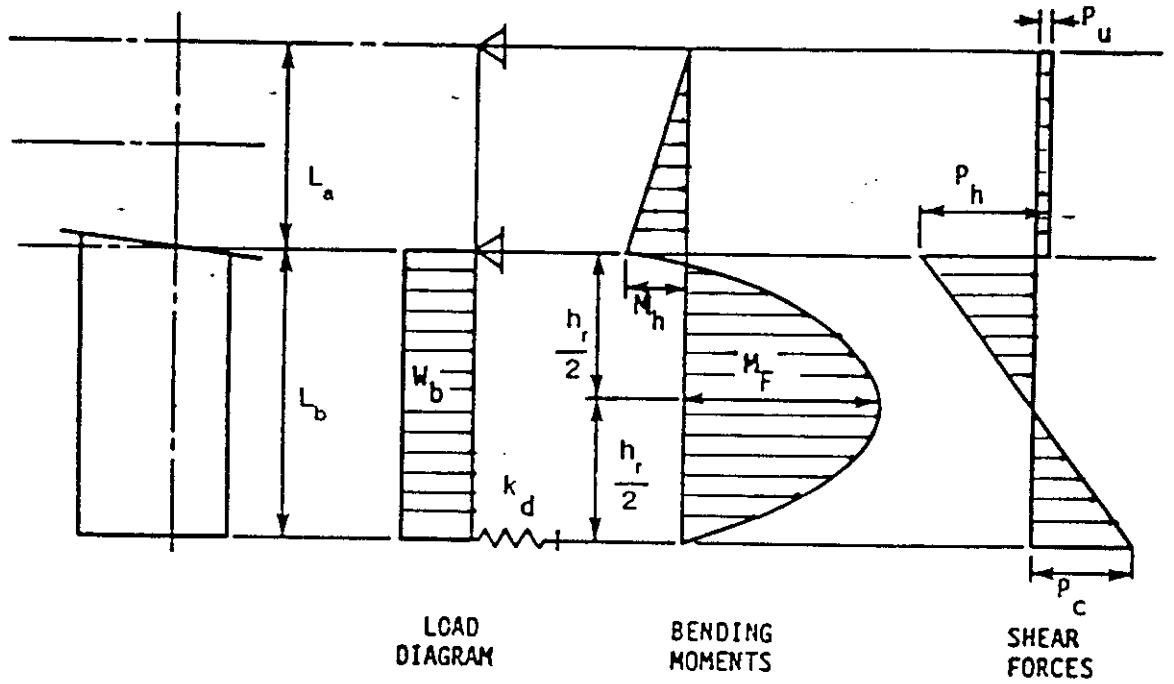


FIGURE 31. *Balanced diving plane spanwise loading, bending moment, and shear distribution – example.*

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APPENDIX A

EXAMPLE 1

SPADE RUDDER CALCULATIONS

10. SCOPE

10.1 General. This appendix presents a numerical example of how calculations for a spade rudder control surface might be carried out. This appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

10.2 Content. After presenting information by hypothesis, this appendix presents sample calculations for the following determinations:

a. Bending moment and shear diagrams, and bearing reactions.

b. Web spacing and web and side plating thicknesses.

c. Stock size.

d. Hub casting wall thickness and taper.

e. Spanwise and chordwise stress.

20. APPLICABLE DOCUMENTS.

20.1 Government documents.

20.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the DODISS and supplement thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-16216	Steel Plate, Alloy, Structural, High Yield Strength (HY-80 and HY-100)
MIL-S-21952	Steel (HY-80 and HY-100) Bars, Alloy
MIL-S-22698	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

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APPENDIX A

MIL-S-23284	Steel Forgings, Carbon and Alloy, for Shafts, Sleeves, Couplings, and Stocks (Rudders and Diving Planes)
MIL-C-24707/1	Castings, Ferrous, for Machinery and Structural Applications

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. HYPOTHESIS

30.1 Information. The following information is given.

Ship category:	Surface ship
Control surface:	Spade rudder, hollow, with dimensions as shown on figure 20.
Torque:	Maximum torque = 12,600 in-kip at maximum design speed
Loadings:	Spanwise and chordwise hydrodynamic load distributions as shown on figure 21.

Load combinations: For overall stress, hydrodynamic load only

Material:	Rudder plate, HY-80 Rudder stock, Steel forging, MIL-S-23284 and as noted in table IV.

Stock bore ratio: $p = d_i/d_o = 0.17$

40. DETERMINATION OF BENDING MOMENT AND SHEAR DIAGRAMS, AND BEARING REACTIONS

40.1 Shear and bending moment diagram and bearing reactions.

40.1.1 Spanwise loading. (See figures 10 and 22.)

a. *Reaction forces.* (See equations (3) and (4).)

$$P_u = \frac{N(z + h_n)}{L_a} \qquad kip$$

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N = Resultant force acting at CP, derivable from the area of spanswise (chordwise) load diagram (see figure 21)

$N = (1/2)(47.4 - 30.9)(21.25) + (30.9)(21.25)$

$N = 832$ kip

z = Spanwise distance to CP, which corresponds to center of area of load diagram

$z = [(1/2)(47.4 - 30.9)(21.25)(2/3)(21.25) + (30.9)(21.25)(21.25/2)]/832$

$z = 11.4$ ft

$h_n = 2.58$ ft (see figure 20)

$L_a = 7.25$ ft (see figure 20)

$$P_u = \frac{832\,(11.4 + 2.58)}{7.25} = 1600 \text{ kip}$$

$P_h = N + P_u = 832 + 1600 = 2432$ kip

b. *Maximum bending moment.* (See equation (5))

$$M_h = N(z + h_n)$$
$$= 832(11.4 + 2.58) = 11{,}600 \; kip\text{-}ft$$

40.1.2 Chordwise loading. (See figure 23.)

$N_R = 452$ kip
$1_R = 4.18$ ft
$N_F = 380$ kip
$1_F = 2.34$ ft

a. Chordwise reaction forces: $P_R = N_R = 452$ kip
$P_F = N_F = 380$ kip

b. Bending moment: $M_R = 1_R N_R = (4.18)(425) = 1890$ kip-ft
$M_F = 1_F N_F = (2.34)(380) = 889$ kip-ft

50. DETERMINATION OF WEB SPACING AND WEB AND SIDE PLATING THICKNESS
(Control surface is hollow as specified.)

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APPENDIX A

50.1 Determining ratio of panel breadth to plating thickness, b/t. (See discussion accompanying equation (22).)

$$\frac{b}{t'} = \frac{C}{K\,H}$$

50.1.1 Establishing b/t'.

a. As a place to start, one may assume that:

b/a = 0.83
K = 0.87 (see table III)
C = 750 $ft^{1/2}$ for HY-80 (see table II)

b. *Determining total head, H.*

H = Total head of seawater (feet)
= $h_1 + h_2$
= Hydrostatic head h_1 due to hydrodynamic pressure plus hydrostatic head h_2 due to draft of ship

h_1 = W_t/(specific weight of seawater)
(Specific wt of seawater = 64.0 lb/ft^3)

W_t = Hydrodynamic pressure (kip/ft^2)
= $[(47.4)(2)]/15.5 = 6.12\ kip/ft^2$

h_1 = 6120/64.0 = 95.6 ft

h_2 = 28 ft (draft of ship)

H = 95.6 + 28 = 124 ft

$$b/t' = \frac{750}{0.87 \quad 124} = 77$$

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APPENDIX A

50.1.2 Plating thickness, t_A. (Not less than 0.5 inch. See 5.6.1.2.)

Assume: t′ = 0.5025 inch (t′ is selected to suit expected load)

$t = t' + t_1 + t_2 = 0.5025 + 0.06 + 0 = 0.5625$ inch (0.06-inch corrosion allowance, 0.0-inch cavitation allowance (see 5.6.2.2).)

50.1.3 Panel breadth and length.

b = (b/t′)t′ = (77) (0.5025) = 39 inches
b/t = 39/0.5625 = 69

a = (39)/(0.83) = 47 inches

60. DETERMINATION OF STOCK SIZE

60.1 Method I. (See equation (20).)

$$d_1 = 3\frac{16n}{\sigma_y \pi(1-P^4)}(M + M^2 + M_t^2)$$

$$= 3\frac{(16)\quad(2)}{(65)(\pi)(1-0.17^4)}(139{,}000 + (139{,}000)^2 + (12{,}600)^2)$$

$$= 35.2 \textit{ inches}$$

60.2 Method 2. (see equation (21).)

$$d_2 = 3\frac{16n}{0.6\pi\sigma_y(1-P^4)}M^2 + M_t^2$$

$$= 3\frac{(16)(2)}{(0.6)(\pi)(65)(1-0.17^4)}(139{,}000)^2 + (12{,}600)^2)$$

$$= 33.2 \textit{ inches}$$

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APPENDIX A

60.3 Comparison and selection. The larger of the two diameters found in 60.1 and 60.2 should be selected, but the stock diameter may have to be increased in order to accommodate standard size bearings. In this particular instance, the next larger bearing size requires a stock diameter of 35.5 inches.

70. DETERMINATION OF HUB CASTING WALL THICKNESS AND TAPER

70.1 Hub casting wall thickness. The narrowest acceptable wall thickness may be determined from the following equations and description.

$$\sigma = KMc/I, \textit{ in which}$$

σ	=	Reduced yield strength of hub material, in kip/in^2, = 0.6 σ_y
σ_y	=	Yield strength of stock material, 65 kip/in^2
K	=	Stress factor, to be found from formula or curve on figure 16
M	=	Bending moment in stock, 139,000 in-kip, in this example
c	=	$s + d_o/2$, inches
s	=	Hub wall thickness (quantity to be solved for in this example)
d_o	=	Outer diameter of stock, 35.5 inches
I	=	Second moment of area of section of stock about a diameter (commonly called moment of inertia of section), in^4
I	=	$(\pi/64)[(d_o + 2S)^4 - d_i^4]$, but since d_i is much less than $(d_o + 2S)$, d_i may be ignored in the difference between fourth powers.
I	=	Approximately $(\pi/64)(d_o + 2s)^4$
I	=	Approximately $(\pi/64)(2c)^4 = (\pi/4)c^4$

Stock penetration into hub is estimated at 35 percent of span of rudder, that is, 35 percent of 21 feet, 3 inches, or 89.25 inches. This penetration of 89.25/35.5 or slightly more than 2.5 stock diameters, is the ratio used in entering figure 16 to find K. From figure 16, K = 1.5.

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APPENDIX A

In view of the foregoing,

$$\sigma = 0.6 \times 65{,}000\ lb/in^2 = 39{,}000\ lb/in^2$$

$$\sigma = \frac{1.5\ Mc}{(\pi/4)\ c^4} = \frac{6.0\ M}{\pi c^3},\ whence$$

$$c^3 = \frac{6.0\ M}{\pi\sigma} = \frac{(6)(139{,}000)\ in\text{-}kip}{\pi\ \ (39)\ kip/in^2} = 6810\ in^3,\ and$$

$$c = 19.0\ inches$$

By the defining equation for c as a function of s, s may be solved for to give

$$s = c - d_o/2 = 19.0 - 35.5/2 = 1.25\ inches$$

But, since this value of hub wall thickness is less than the specified lower limit of 15 percent of the stock diameter, the wall thickness needs to be increased to at least 0.15 times 35.5 or 5.32 inches.

70.2 Stock taper. For practical reasons of manufacture, it is proposed that the diameter of the stock at its end within the rudder hub be 14 inches less than its diameter where it enters the rudder hub. The taper in inches per foot is calculated as follows:

$$\frac{Difference\ in\ diameters}{Length\ of\ taper} = \frac{14\ inches}{(89.25\ in)(1\ ft/12\ in)} = 1.9\ in/ft$$

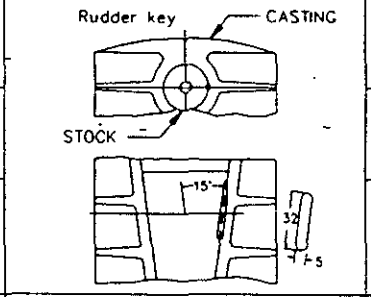
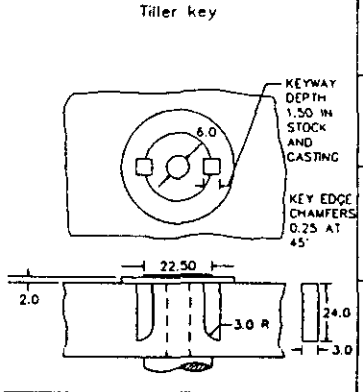
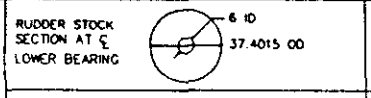
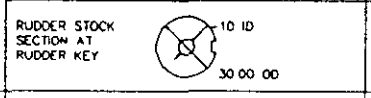
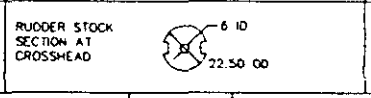
80. SPANWISE AND CHORDWISE STRESS

(See table IV for further numerical relationships.)

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APPENDIX A

Table IV. *Stress information.*

Material (steel)	Section shape (lengths in inches)	Section area (in^2)	Section modulus (in^3)	Kind of Stress	Maximum load or bending moment	Maximum combined stress (lb/in^2)	Percent of yield point
Steel plate MIL-S-16216 HY80 YP 80,000 lb/in^2	Chordwise section of rudder	-----	3,200	Bending	49,500 kip-in	15,500	19.4
	Spanwise section of rudder	-----	30,774	Bending	22,600 kip-in	734	0.918
	Spanwise section of rudder	-----	5,432	Bending	10,200 kip-in	1,880	2.35
MIL-S-21952 TY-B HY80 YP 48,000 lb/in^2		154.63		Shear	800,000 lb	5,170	10.8
MIL-S-21952 TY-B HY80 YP 128,000 lb/in^2		(27)(1.75) = 47.3		Bearing on stock and key	857,000 lb	18,200	14.2
Casting MIL-C-24707/1 Class B YP 128,000 lb/in^2		(27)(1.75) = 47.3		Bearing of key on rudder casting	750,000 lb	15,900	12.4
MIL-S-21952 TY-B HY80 YP 48,000 lb/in^2		2 (3) (24) = 144		Shear	1,043,000 lb	7,240	15.1
MIL-S-21952 TY-B HY80 YP 128,000 lb/in^2		2 (1.25) (22.5) = 56.3		Bearing on stock and key	1,170,000 lb	20,800	16.3
Casting MIL-C-24707/1 Class B YP 128,000 lb/in^2		2 (1.25) (24.0) = 60.0		Bearing of key on tiller casting	942,000 lb	15,700	12.3
Retainer ring MIL-S-22698 YP 52,800 lb/in^2		(22.5 π) (2) = 141		Shear	165,000 lb	1,170	2.2
Steel forging (rudder stock) MIL-S-23284 YP 65,000 lb/in^2			10,300 5,130	Torsion Bending Combined	12,000 kip-in 139,300 kip-in	1,170 27,200 27,300	42.0
			5,240 2,620	Torsion Bending Combined	12,000 kip-in 84,720 kip-in	2,290 32,300 32,500	50.0
MIL-S-23284 YP 39,000 lb/in^2			2,230	Torsion	12,000 kip-in	5,380	13.8

Load stress limits	
	Thrust load on upper bearing: Not to exceed 165,000 lb Radial load on upper bearing: Not to exceed 1,625,000 lb Radial load on lower bearing: Not to exceed 2,452,000 lb

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APPENDIX B

EXAMPLE 2

HORN RUDDER CALCULATIONS

10. SCOPE

10.1 General. This appendix presents a numerical example of how calculations for a horn rudder control surface might be carried out. This appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

10.2 Content. After presenting information by hypothesis, this appendix presents sample calculations for the following determinations:

a. Bending moment and shear diagrams, and bearing reactions.
b. Gudgeon, pintle, and keyway calculations.
c. Deflections of rudder stock.

10.3 Commonality with spade rudders. The following determinations are omitted from this appendix because they follow the same procedures as described for spade rudders in example 1 of appendix A:

a. Web spacing and web and side plating thicknesses
b. Stock size
c. Hub casting
d. Spanwise and chordwise stress

20. APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

30. HYPOTHESIS

30.1 Information. The following information is given:

Ship category:	Surface ship
Control surface:	Horn rudder, hollow, with dimensions as shown on figure 24.
Torque:	Maximum torque = 16,135 in-kip at maximum design speed

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APPENDIX B

Loadings: Spanwise hydrodynamic load distributions as shown on figure 25. (Only the spanwise loading is shown for this example. In practice the chordwise loading would also be given, much as for example 1 in appendix A.)

Load combinations: For overall stress, hydrodynamic load only.

Material: Rudder plate, HY-80
Rudder stock, Steel forging, MIL-S-23284 (see section 20 of appendix A).

40. DETERMINATION OF BENDING MOMENT AND SHEAR DIAGRAMS, AND BEARING REACTIONS

40.1 Shear and bending moment diagrams and bearing reactions (spanwise loading only). Spanwise loading is as shown on figure 26, in which

$N = 1702$ kip
$I_a = 27{,}000$ in^4 (estimated)
$I_b = 160{,}000$ in^4 (estimated)
$I_h = 760{,}000$ in^4 (estimated)
$J_h = 37{,}000$ in^4 (estimated)

40.1.1 Maximum bending moments. (See figures 25 and 26 and equations (8) through (14); see also figure 12.) The following dimensions will be used in the calculations:

$h_n = 1.14$ ft
$L_a = 3.77$ ft
$L_b = 11.61$ ft
$L_c = 13.63$ ft
$L_h = 9.91$ ft
$m = 3.55$ ft

a. At pintle bearing.

$$M_p = \frac{(W_c)(L_c^2)}{2} = \frac{(99.5)\ (13.63)^2}{2} = 9240\ ft\text{-}kip$$

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$$K = \frac{E}{L_h\left(\frac{2.6m^2}{J_h} + \frac{L_h^2}{3I_h}\right)} = \frac{E}{(9.91)\left[\frac{(2.6)(3.55)^2}{(37{,}000)/(12)^4} + \frac{(9.91)^2}{3\ (760{,}000)/(12)^4}\right]}$$

$$= \frac{E}{190}\ kip/ft$$

$$a = \frac{KL_b^3}{3EI_b} = \frac{\frac{E}{190}\ (11.61)^3}{(3)(E)((160{,}000)/(12)^4)} = 0.36\ (dimensionless)$$

b. At neck bearing (also known as middle bearing).

$$M_h = \frac{W_b(L_b^2 - h_n^2)\left(\frac{1}{2} + \frac{a}{8}\right) + W_cL_c^2\left(\frac{1}{2} + \frac{L_b}{L_c} - \frac{a}{4}\right)}{1 + a\ \left(1 + \frac{4L_aI_b}{3L_bI_a}\right)}$$

$$= \frac{(33.1)(11.61^2 - 1.14^2)\left(\frac{1}{2} + \frac{0.36}{8}\right) + (99.5)(13.63)^2\left(\frac{1}{2} + \frac{11.61}{13.63} - \frac{0.36}{4}\right)}{1 + (0.36)\ \left(1 + \frac{(4)(3.77)(160{,}000)/(12)^4}{(3)(11.61)(27{,}000)/(12)^4}\right)}$$

$$= 11{,}300\ kip\text{-}ft$$

40.1.2 Bearing reaction forces.

$$P_u = \frac{M_h}{L_a} = (11{,}300)/(3.77) = 3000\ kip$$

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$$P_h = P_u \left(1 + \frac{L_a}{L_b}\right) + \frac{W_b}{2\,L_b}(L_b - h_n)^2 - \frac{M_p}{L_b}$$

$$= 3000\left(1 + \frac{3.77}{11.61}\right) + \frac{33.1}{2(11.61)}(11.61 - 1.14)^2 - \frac{9240}{11.61} = 3330\ kip$$

$$P_p = N + P_u - P_h = 1702 + 3000 - 3330 = 1370\ kip$$

40.1.3 Control surface horn. (See figures 13 and 27.)

Shear force = P_p = 1370 kip (see 40.1.2).

Maximum bending moment = $(P_p)(L_h)$ = (1370)(9.91) = 13,600 ft-kip

Torque = $(P_p)(m)$ = (1370)(3.55) = 4860 ft-kip

50. GUDGEON, PINTLE, AND KEYWAY CALCULATIONS

50.1. Gudgeons. (See equation (26).)

R_1 = 27.125 inches
R_2 = 15.6875 inches
a = $(R_1 + R_2)/20$ = 2.14 inches
L = 3 ft, 3 inches = 39 inches
T = 1370 kip
G = 3.73 (For R_1/R_2 = 1.73, from figure 17)

$$\sigma_G = \frac{G\,T}{(10a)L} = \frac{(3.73)(1370)}{(10)(2.14)(39)} = 6.1\ kip/in^2$$

50.2 Pintles. (For pintles see example 3 of appendix C.)

50.3 Keyways. (A numerical example is not presented. See 5.9.3.)

60. CALCULATION OF STOCK DEFLECTION

60.1 Objective. To find the stock deflection at a point 6 inches below the lower bearing. (See 5.4 and figure 12.)

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APPENDIX B

60.2 Finding bending moment. At point B on figure 26 the bending moment, M_h, is 11,300 kip-ft or 1.3662E8 in·lb. At point X, a distance x from point A, the moment is

$$M_x + (x/45.24)(1.36E8)\ in{\cdot}lb \quad (x \leq \overline{AB})$$

Between points B and X_s the shear is

$$P_h - P_u = 3330 - 3000 = 330\ kip$$
$$= 3.30E5\ lb$$

Thus, the moment between points B and X_s may be written

$$M_x = 1.36E8 - (3.30E5)(x-B)\ in{\cdot}lb \quad (x \geq \overline{AB})$$

Since the moment varies linearly between these points, only the moments at points A, B, and X_s need be tabulated.

Point designator	x (inches)	M (in·lb)
A	0.0	0.0
B	45.24	1.36E8
X_s	51.24	1.34E8

60.3 Determining moment of inertia of cross-section of stock. For present purposes the moment of inertia, $I=I_a$, may be taken as a constant as given, that is,

$$I = 27{,}000\ in^4$$

60.4 Calculating M/EI ratios. The modulus of elasticity, E, is a property of the material used.

$$E = 2.9E7\ lb/in^2$$

From this modulus, the moment of inertia from 60.3, and the three values of M from 60.2, the following set of M/EI diagram coordinates may be developed:

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Point designator	x (inches)	M/EI (inches $^{-1}$)
A	0.0	0.0
B	45.24	1.74E-4
X_s	51.24	1.72E-4

These points are plotted on figure 28.

60.5 Calculating deflections. For x > $\overline{AB}$, the deflection may be found from (see 5.4):

$$v_x = t_{xb} + \left[(x-B)/(B-A) \right] t_{AB}$$

t_{xB} = (the definite integral of M/EI with respect to x from B to X_s) times (the abscissal interval from X_s to the centroid of the area that represents this definite integral)

"area" = (1.72E-4)(6) + (0.5)(1.74E-4 - 1.72E-4)(6)
= 1.03E-3 + 9.OOE-6
= 1.04E-3 (dimensionless)

abscissal interval = (1.03E-3)(3)/(1.04E-3)
+ (4)(9.00E-6)/(1.04E-3) = 3.00 inches

Hence,

t_{xB} = (1.04E-3)(3.00) = 3.12E-3 inch

$(X_s - B)/(B-A) = 6/45.24 = 0.1326$

Also,

t_{AB} = "area" times abscissal interval to centroid
= (0.5)(1.74E-4)(45.24)(2/3)(45.24)
= 0.119 inch

Finally, the deflection at the seal = v_x = 3.12E-3 + (0.1326)(0.1191)
= 0.0189 inch

If the moment varied nonlinearly or if the moment of inertia were not constant, the M/(EI) curve could be determined at a series of x locations, and the tangential deviations evaluated at each.

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APPENDIX C

EXAMPLE 3

DIVING PLANE CALCULATIONS

10. SCOPE

10.1 General. This appendix presents a numerical example of how calculations for a diving plane control surface might be carried out. This appendix is not a mandatory part of the standard. The information contained herein is intended for guidance only.

10.2 Content. After presenting information by hypothesis, this appendix presents sample calculations for the following determinations:

a. Bending moment and shear diagrams, and bearing reactions
b. Web spacing and web and side plating thicknesses
c. Pintles

10.3 Commonality with spade rudders. The following determinations are omitted from this appendix because they follow the same procedures as described for spade rudders in example 1 of appendix A:

a. Stock size
b. Hub casting
c. Spanwise and chordwise stress

20. APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

30. HYPOTHESIS

30.1 Information. The following information is given.

Ship category:	Submarine
Control surface:	Balance diving plane, filled, with dimensions as shown on figure 29.
Torque:	Maximum torque (hydrodynamic) = 938 in-kip at maximum design speed Maximum torque (sea-slap) = 1720 in-kip

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APPENDIX C

Loadings:	Spanwise and chordwise sea slap loads, hydrodynamic load, and ice breakthrough loads. (Of these, only sea slap loads are considered in this example, and chordwise load distribution is omitted.) Only the spanwise loading distribution is shown on figure 30.
Load combinations:	For overall stress, sea slap only. (Combinations of other loadings listed under "loadings" above are not considered in this example, but should be considered in an actual analysis.)
Materials:	Diving plane, HY-80 Stock, HY-80

40. DETERMINATION OF BENDING MOMENT AND SHEAR DIAGRAMS, AND BEARING REACTIONS

40.1 Shear and bending moment diagram. Pertinent parameters are shown on figure 31, where,

$N = 110$ kip
$I_a = 850$ in^4 (estimated)
$I_b = 1600$ in^4 (estimated)
$I_d =$ (unknown but estimated as very large),

and where L_d, h_n, and h_d all equal approximately zero (see also figure 14).

40.2 Bending moments. (See equations (18) and (19).)

$a = \dfrac{L_b^3 I_d}{L_d^3 I_b}$ very large, since I_d and L_b are very much larger than I_b and L_d respectively.

$$M_h = \frac{W_b\,(L_b^2 - h_n^2)(\frac{1}{2} + \frac{a}{8})}{1 + a(1 + \frac{L_d I_b}{L_b I_a})} = \frac{\frac{1}{8} W_b\,(L_b^2 - h_n^2)}{1 + \frac{L_d I_b}{L_b I_a}}$$

approximately, after division of numerator and denominator by a.

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$$= \frac{(\frac{1}{8})(7.9)((13.82)^2 - 0)}{1 + \frac{(10.2)((1{,}600)/(12)^4)}{(13.82)((850)/(12)^4)}} = 79 \; ft\text{-}kip$$

$$M_F = \frac{W_b h_r^2}{8} = \frac{(7.9)(13.82)^2}{8} = 190 \; ft\text{-}kip$$

40.3 Bearing reaction forces. (See equations (15) through (17).)

$$P_u = \frac{M_h}{L_a} = \frac{79}{10.2} = 7.7 \; kip$$

$$P_h = P_u(1 + \frac{L_a}{L_b}) + \frac{N}{L_b}(\frac{h_r}{2} + h_d$$

$$= (7.7)(1 + \frac{10.2}{13.82}) + (\frac{110}{13.82})(\frac{13.82}{2} + 0)$$

$$= 68 \; kip$$

$$P_c = N + P_u - P_h = 110 + 7.7 - 68 = 50 \; kip$$

The calculation has been performed for only one stern plane. Because both stern planes are attached to the same stock, the shears and moments from both are to be superposed in way of the stock between bearings.

50. DETERMINATION OF WEB SPACING AND WEB AND SIDE PLATING THICKNESSES

50.1 Panel breadth-to-thickness ratio, b/t. (See 5.6.2.1)

Chosen: b/t = 72

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50.2 Plating thickness. (See equation (24).)

$t = t' + t_1 + t_2$
$t' = 0.19$ inch
$t_1 = 0.06$ inch (corrosion allowance)
$t_2 = 0$ (cavitation allowance)
$t = 0.25$ inch, which is the minimum acceptable thickness for filled control surfaces (see 5.6.3.3(b)).

50.3 Panel breadth and panel length.

$b = (b/t)t = (72)(0.25) = 18$ inches
a should approximate b; hence a too may equal 18 inches.

60. PINTLES

$P_c = 50$ kip (shear force at pintle)
Sectional area of pintle = 23.5 in^2
Shear stress $= \frac{50}{23.5} = 2.1$ kip/in^2

Yield stress = 24 kip/in^2
Factor of Safety $= \frac{24}{2.1} = 11$, which is acceptable (see 5.9.2).

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	MIL-STD-2189(SH)SECTION 562-2	920602

3. DOCUMENT TITLE
DESIGN METHODS FOR NAVAL SHIPBOARD SYSTEMS SECTION 562-2 CONTROL SURFACE STRUCTURE DESIGN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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